Substituting Reference Solvent σ-Profiles for Learned Ones Degrades COSMO-SAC Solubility Prediction

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Abstract

Physics-informed models route a prediction through a quantity that can be looked up, so supplying the tabulated value ought to improve them. We test that assumption for solid–liquid solubility, where a graph neural network predicts the molecular charge-density profiles a fixed COSMO-SAC layer consumes and quantum chemistry tabulates independently. Grounding names two opposite operations, and they do not behave alike. Substituting the tabulated profile for the learned one at prediction time makes solubility prediction worse at every seed, and degrades the ranking of solvents for a solute, which is the decision the model is used for. Supervising the same network against the same database during training carries no established sign. The operation that fails is the one available in practice, because it needs no retraining. Infinite-dilution activity data separate a deployed thermodynamic model's own error from what its inputs leave unresolved, and the model's error dominates on one solvent chemistry. A search over strata identifies that cell, and an independent set reproduces it at half its size without meeting this paper's own stability rule, so we report it as the instrument's output and not as an established effect. Where the fixed model is the binding constraint, a more accurate input exposes its misspecification instead of repairing it.

1 Introduction

How much of a solid dissolves in a given liquid decides whether a drug is formulated as a tablet or an injection, which solvent a crystallisation runs in, and how a product is purified. Each value is measured one solute, one solvent and one temperature at a time, so most industrially relevant combinations remain unmeasured. Independent determinations of the same system disagree by enough to floor what any predictor can achieve. Dissolving a solid also trades breaking a crystal against mixing the freed molecules into the solvent, so a predictor must be right about two unrelated kinds of chemistry at once, on molecules unlike any it was shown.

Predictors divide by how much physical structure they impose. Direct models regress solubility from descriptors or from a representation learned off the molecular graph; FastSolv[1] does so with a deep neural network and leads on extrapolation to new solutes. They are the more accurate there and they are opaque, the prediction arriving as a single number, silent about which part of the chemistry it captured. Physics-informed models instead predict the quantities the physical picture names—an activity coefficient for the solute–solvent interaction, a melting temperature and heat of fusion for the crystal—and combine them through a fixed equation. SolProp[2] does so with a thermodynamic cycle of separately learned components. The fixed equation buys decomposability: each part is a quantity chemists tabulate on its own, so it can be supervised against reference data a direct regressor cannot use. On accuracy the black box leads and the physics-informed cycle trails it. Learned surrogates for the activity coefficient split the same way. Winter et al.[3] predict $\ln \gamma_2^\infty$ from SMILES strings, the line notation for molecular structure, with a language model; Sanchez-Medina et al.[4] and Wang et al.[5] embed a Gibbs–Helmholtz constraint in a graph network for its temperature dependence. All three make the activity coefficient the regression target. A second line keeps a mechanistic model and learns its *input*. COSMO-SAC[6] is a segment-activity reformulation of Klamt's conductor-like screening model for real solvents (COSMO-RS), and the σ-profile it consumes—the screening-charge density over a molecule's cavity—is that framework's construct, tabulated from quantum chemistry.[7] Predicting it from structure removes that calculation from the pipeline. Group-contribution[8,9] and then learned predictors[10–12] have supplied it in place of the calculation. How far the tabulated reference itself moves with the quantum chemistry and solvation treatment behind it is its own question.[13] Islam and Chen[14] instead regress an *apparent* profile from phase-equilibrium data, the pre-neural form of the effective profile this work measures. TeNNet-SAC[15] does so while also learning the segment-activity kernel and the cavity geometry, and reports accuracy better than COSMO-SAC's after end-to-end finetuning. Its headline is the opposite of this one's, and the two score different operations. Its gain is a training-time gain, which over five leak-free seeds has no established sign here. Its externally supervised profile predictor then stays frozen while only the output head is finetuned, so the closure's gradient never reaches the intermediate. This work's negative result is about the inference-time substitution, which TeNNet-SAC leaves unreported, on an intermediate trained through the closure with neither safeguard.

A third design removes the intermediate. HANNA[16] parameterises the excess Gibbs energy directly from SMILES with no quantum chemistry, hard-coding the pure-component limit, the Gibbs–Duhem relation and permutation symmetry in the architecture rather than the loss. It reports higher accuracy than UNIFAC, the group-contribution activity-coefficient method standard in process design, over 317,421 mixture points. The two published designs bracket what this work measures: admissibility can be bought with quantum labels or built into the functional form, and this model does neither.

When a fixed physical structure improves or degrades a learned model is the broader question of physics-informed machine learning.[17] Its dominant forms impose structure as a differentiable object: a governing equation enforced through the loss,[18] gray-box hybrids composing learned components with fixed mechanistic maps,[19] and operator learning with a trainable correction to a misspecified equation.[20] More physics is expected to improve generalisation, yet the same structure has documented failure modes in which it makes optimisation or accuracy worse.[21] In neural solvers for partial differential equations a learnable term added to a fixed discretised operator absorbs truncation artifacts instead of physics, so the apparent interpretability is illusory.[22] The error source there is numerical, which puts it beyond external reference. The same concern appears in *concept-bottleneck models*,[23] networks routed through a layer of named, human-interpretable concepts that are the whole of what the prediction is made from. Concept bottlenecks have the shape of a learned intermediate feeding a fixed physical map, and there the failure is *concept leakage*: a learned concept silently encodes information beyond its nominal meaning.[24–26] One route to it runs through the downstream model: a misspecified head forces the encoder to deform the concept into carrying a dependence beyond that head's reach.[27] Overwriting a frozen head's leaked concepts with ground truth can then degrade accuracy.[28] The penalty is older than that literature. Koh et al.[23] already report it: in a joint bottleneck trained to prioritise label accuracy over concept accuracy, replacing the predicted concepts with the true ones at test time slightly increases test error. The gap between a fixed physical map and reality is the model-discrepancy problem of Kennedy and O'Hagan[29] and its calibration critique.[30] Four further literatures—forecast decomposition, quasi-maximum likelihood, identifiability, and the validity of composed pipelines—supply pieces we use below.

The design rests on an assumption rarely stated because it is taken for self-evident: a model supplied with physical inputs closer to their true values predicts better. It needs testing, and for one class of intermediate it can be tested outright, because a reference value is tabulated and can be handed to the model in place of the one it computed. When a predictor of this shape is inaccurate, either the fixed physical model is wrong or the learned quantities feeding it are uninformative, and the two call for opposite remedies. The literature lacks a way to *measure*, rather than diagnose, which of the two limits accuracy, and whether the learned intermediate has stopped reporting the quantity it is named after. In the concept-bottleneck case the external reference that would settle either question is missing. We supply one—a reference σ-profile for solubility, tabulated substituent constants for a second chemical domain—and hold the mechanistic model fixed, so the tabulated value stays available as an external reference for the learned one.

Here a graph neural network predicts the pair of charge-density distributions COSMO-SAC consumes, in place of the quantum-chemical calculation that ordinarily supplies them. The tabulated counterpart enters at either of two points: as a target the network is trained toward, or as a replacement for what the network computed at the moment of prediction (Fig. 1). Those are the two operations the word *grounding* names, and only one of them carries a sign. Training toward the reference moves the error four ways out of five and the fifth the other way; supplying it at the moment of prediction raises the error at every seed. Tabulated profiles serve a third purpose on activity-coefficient data, where they divide the error of a fixed model between the model itself and the inputs handed to it. Solubility withholds that division, a solubility measurement constraining the crystal and the mixing contributions jointly and leaving the blame between them unassigned. The division returns a map and not a verdict: one answer per stratum, a stratum being a chemically defined set of rows. The chemistry the model is asked about decides whether a fixed model's own error stands above what its inputs fail to carry. In the strata where it does stand above, a more accurate input exposes the model's bias instead of removing it. The learned distribution meanwhile drifts from its tabulated counterpart along a direction shared across molecules, and stops reporting the quantity it is named after. Tabulated substituent constants behave the same way inside a fixed Hammett relation, the linear equation predicting how readily an acid gives up its proton from the constants tabulated for its substituents.

The article is organised as follows. Section 2 defines the composed predictor, the decomposition of its error into a part belonging to the fixed physical model and a part belonging to the inputs handed to it, and the data, training and evaluation protocols. Section 3 reports the two grounding operations on solubility, what substitution costs in accuracy and in the ranking of solvents, how the learned intermediate departs from its tabulated counterpart, and the error decomposition on activity-coefficient data for two fixed models. Section 4 states the result and its scope.

This work does not propose a solubility model and claims no accuracy for the one it uses. The model is the instrument in which a property of a fixed thermodynamic closure is measured, and every result below is a contrast between two readings of one trained network rather than a comparison between networks.

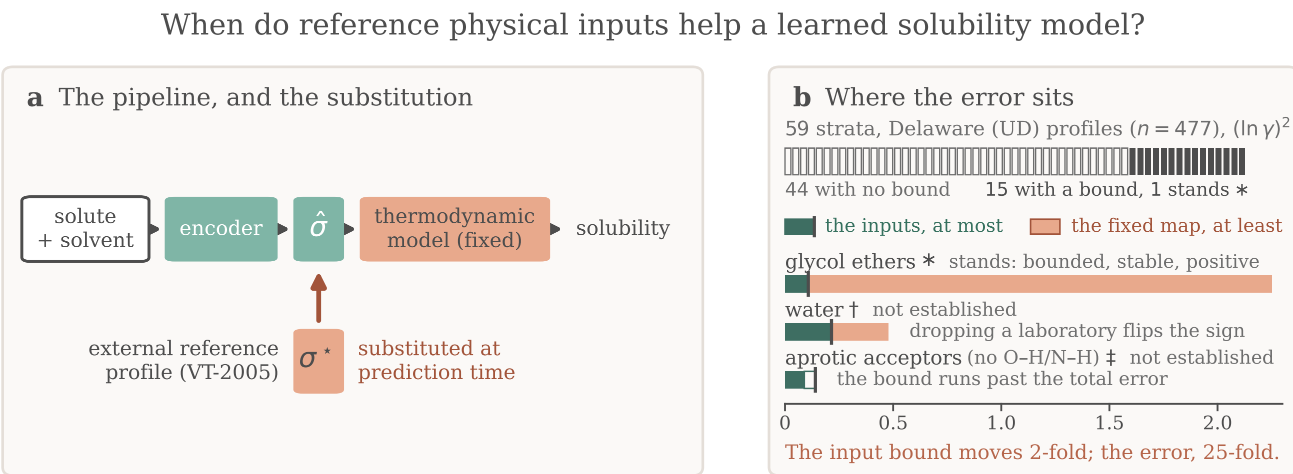


Figure 1: **Overview.** **(a)** A learned encoder predicts a charge-density profile $\hat{\sigma}$ that a fixed thermodynamic model turns into a solubility, and the arrow marks the substitution of the reference $\sigma^\star$ for it. Training-time supervision is omitted. **(b)** The counterpart error on infinite-dilution activity data: the strip is every stratum searched, the bars the three solvent classes with a bound, and the star the one class the admissibility rule leaves standing. Each block is a one-sided bound, not an interval.

2 Computational methods

2.1 The composed predictor and its error split

For a solute s in solvent v at temperature T, the target is $y = \ln x_2$, the log saturated mole fraction. Dissolving a solid trades breaking the crystal lattice (Φ) against mixing the freed molecules into the solvent ($\ln\gamma_2$), and solid–liquid equilibrium (SLE) sets the log-solubility to their negative sum:

$$\ln x_2 = -\underbrace{\frac{\Delta H_{\text{fus}}}{R}\left(\frac{1}{T} - \frac{1}{T_m}\right)}_{\Phi(T)\ \text{(crystal)}} - \underbrace{\ln\gamma_2(x_2, T)}_{\text{activity}}, \quad (1)$$

with no constant-ΔC_p correction (ΔC_p being the difference in heat capacity between the liquid and the solid). That term is absent from the displayed equation because it is absent from the deployed model, which fixes $\Delta C_p = 0$, the head that would predict it being off in the pinned configuration. The exposure is one-signed and substantial: at the group-contribution value this corpus carries, $60\,\mathrm{J\,mol^{-1}\,K^{-1}}$, the omitted term is a median 0.32 in $\ln x_2$ over the 1318 labelled test rows carrying a measured T_m, positive on every one of the 31 solutes—the size of the substitution penalty of §3.1. It enters the physics arms, leaves the closure-free control untouched, and cancels exactly in the evaluation-time substitution, where Φ is common to the two arms (§S1). It survives into the training of the σ head, fitted through a crystal term carrying a bias of that size and direction, whose consequence §3.3 reads. The activity term is produced by a *differentiable closure* g. A graph encoder predicts a σ-profile (the distribution of screening charge density over a molecule, the histogram a COSMO model consumes) for each molecule, and COSMO-SAC,[6] a segment-activity reformulation of COSMO-RS, maps the pair of profiles to $\ln\gamma_2$. The segment-interaction energy inside that map is the closure's *kernel*: the electrostatic-misfit and hydrogen-bonding terms setting what two surface segments cost each other, and the part the published revisions rewrite. A closure variant is therefore named for its kernel's year — the *2002 kernel* this pipeline runs, and the *2010 kernel* that types segments as hydrogen-bond donors or acceptors instead of treating hydrogen bonding with one parameter. Write $z^\star$ for the *reference* latent, a tabulated σ-profile, and $\hat{z}$ for the head's own prediction. **Two profile databases supply $z^\star$ below and they are**

never interchangeable: *VT-2005*, the Virginia Tech 2005 tabulation,[7] which the σ head is supervised against and which is substituted for $\hat{z}$ at prediction time, and *UD*, the University of Delaware successor, which the activity-coefficient rows of §3.5.1 are matched to. Every set, margin and figure below names which it is computed on, and we read no quantity across them; the one place they are deliberately set against each other is the database-to-database bracket of §3.5.3. Feeding $z^\star$ in place of $\hat{z}$ (a σ-oracle) isolates the closure from encoder error: were the closure well specified, $g(z^\star)$ would be the best available activity estimate. Each trained configuration, and each closure variant evaluated, is an *arm*; a *DirectGNN* control arm drops the closure and emits $\ln x_2$ directly, and we tuned the two separately (§2.4).

We state everything below for a general *composed predictor* $\hat{y} = g(h(x))$ in which a fixed closure g consumes a learned physical intermediate $z = h(x)$, against a matched free predictor that shares the encoder and drops g. Solubility is one instance, and a second is the closure of §3.6 for pK_a, the acid dissociation constant on a negative logarithmic scale.

Solubility alone underdetermines the crystal/activity split of Eq. (1). At infinite dilution with $\Delta C_p = 0$ the observable is affine in $1/T$, and the activity intercept and slope already span that plane, so the profiled Fisher information for ΔH_{fus} is exactly zero and the indeterminacy set is two-dimensional. At finite dilution the composition factor $(1 - x_2)^2$ leaves an information of order x_2^2. The deficiency closes under a rank-completing external constraint and nothing weaker: a direct crystal label, or a fixed activity slope (Lemma S1, §S2). That is why an under-determined factor needs external single-component data and why we expect compensation between the factors, and it is silent on whether grounding helps.

2.1.1 The closure–insufficiency decomposition

Let m be the closure's target (here the experimental infinite-dilution activity coefficient (IDAC) $\ln \gamma_2^\infty$, in §3.6 the measured pK_a) and $g(z^\star)$ the closure evaluated on the reference latent.

Lemma 1 (Exact closure–insufficiency decomposition and its one-sided bounds)**.** *Let g be a* fixed *(non-fitted) closure and let $z^\star$ denote its* entire *argument (the profile pair together with the temperature wherever the closure reads one). Then* (a) *the reference-input mean squared error splits exactly,*

$$\underbrace{\mathbb{E}\big[(m - g(z^\star))^2\big]}_{\textit{ref.-input MSE}} = \underbrace{\mathbb{E}\big[\mathrm{Var}(m \mid z^\star)\big]}_{B_{\text{insuff}}\ (\textit{irreducible})} + \underbrace{\mathbb{E}\big[(\mathbb{E}[m \mid z^\star] - g(z^\star))^2\big]}_{B_{\text{closure}}\ (\textit{decoder bias})}, \tag{2}$$

the cross term vanishing identically; (b) $B_{\text{closure}} \geq (\mathbb{E}[m] - \mathbb{E}[g])^2$*;* (c) $B_{\text{insuff}} \leq \mathbb{E}[\mathrm{Var}(m \mid \mathrm{bin}(g(z^\star)))]$ *for any binning of $g(z^\star)$; and* (d) B_{insuff} *is a functional of the law of $(m, z^\star)$ alone and does not involve g, so it is identical across closure conventions, while the bound in (c) is not.*

The proof is in §S2. Two of its parts are conditions of use rather than steps, and both are stated here. **The identity holds because $g(z^\star)$ is $\sigma(z^\star)$-measurable, and it fails if any argument of g is left out of the conditioning variable**: with $g = g(z^\star, T)$ and T excluded, the cross term need not vanish and the binning stops being a coarsening of the conditioning σ-algebra, which takes the identity and the upper bound with it. Where g reads a temperature (the broad IDAC set of §3.5.1, dim $=$ 103) we accordingly take $z^\star = (\text{profile pair}, T)$. On the VT-2005-matched set every row sits at 298 K and the two readings coincide. The Hammett closure of §3.6 reads a per-series constant pair as well as the substituent constant, so there $z^\star = (s, \sigma_{\text{H}}^\star)$ and we condition on both.
B_{closure} is the systematic discrepancy of a fixed physical map in the sense of Kennedy–O'Hagan,[29,30] and its dominance over B_{insuff} is the empirical content of the claim that the ceiling is set by the closure and not by the inputs. B_{closure} **is the error of a composite object.** The identity compares $g(z^\star)$ against $\mathbb{E}[m \mid z^\star]$, and $z^\star$ is produced by a procedure:

for σ-profiles, a quantum-chemical calculation on one conformer and one protonation state per molecule at one level of theory. B_{closure} is therefore a joint property of the map and of that procedure, not of the kernel's functional form, and a displaced reference loads onto it through the curvature of g even where g is exactly specified (§S8). The deployed pipeline consumes that same composite, so every attribution below is to the pair. Where model-discrepancy calibration *fits* a correction from held-out observations, here g is fixed and $z^\star$ supplied from outside the fit, so we *attribute* the composite's error instead, which is available only where an external latent oracle exists. **Exactness is contingent on g being fixed**: for a *fitted* decoder the cross term survives and a plug-in point split is invalid.

Because B_{insuff} is bounded above and B_{closure} below, a positive *separation margin* $\text{MSE} - 2B^{\text{up}}_{\text{insuff}}$ establishes that the closure's own error exceeds what input insufficiency can account for. A margin at or below zero fails to separate and licenses nothing: a reversal would need a *lower* bound on B_{insuff}, which lies beyond this instrument. **Its size is a lower bound and not an estimate.** Since $\text{MSE} = B_{\text{insuff}} + B_{\text{closure}}$ exactly, $B_{\text{closure}} - B_{\text{insuff}} = \text{MSE} - 2B_{\text{insuff}}$, so the reported margin understates that difference by $2(B^{\text{up}}_{\text{insuff}} - B_{\text{insuff}}) \geq 0$, an amount beyond this design's reach. Every margin below therefore says the closure's error exceeds its inputs' by *at least* the number printed, and a bootstrap interval on a margin bounds that bound, its upper endpoint limiting nothing.

What population, and what sets the resolution. The population is the *measurement superpopulation*: a draw is a nominal compound pair at a temperature, with the conformer, tautomer and protonation state actually populated there, and an independent determination of $\ln \gamma_2^\infty$ for it. The conditioning variable holds the reference profile and the temperature and leaves the conformational state and the replicate outside, so $\text{Var}(m \mid z^\star)$ is the spread of $\ln \gamma_2^\infty$ across the conformational ensemble and across replicate determinations at one profile and temperature. That spread is strictly positive for physical reasons, and experimental label noise is a *component* of it. *Observing* it is what the design forbids, each set holding at most one row at a given argument. The bounds are therefore conservative in one direction alone: were the superpopulation variance negligible, the finding would be the stronger $B_{\text{closure}} = \text{MSE}$. So overestimating B_{insuff} produces no artifact in what is reported, every entry is a bound and not the inputs' share of the error, and a low cluster-bootstrap frequency P is loss of resolution and not evidence for the inputs. Resolution is governed by $\dim(z^\star)/n$.

2.1.2 The decomposition and the grounding gap

The decomposition is the measurement the conclusions rest on: assumption-light, needing neither a representable encoder nor a lossless bottleneck. Beside it we report a directly observable quantity, the *grounding gap* $\mathcal{G} := R_{\text{orc}} - R_{\text{e2e}}$, the oracle risk minus the best end-to-end risk; the paradox is $\mathcal{G} > 0$. Under a representable encoder (A1) and a lossless bottleneck (A2) it is sandwiched $0 \leq \mathcal{G} \leq B_{\text{closure}}$, so $\mathcal{G} > 0$ would certify $B_{\text{closure}} > 0$ (Theorem S1, §S2). A1 is unverified: the linear probe of §S6.1 scores an untrained encoder of the same architecture within its own spread, so it does not bear on A1. A2 is unverifiable and physically dubious, and when it fails $\mathcal{G} > 0$ conflates closure misspecification with input insufficiency, a well-specified closure with a lossy latent giving $B_{\text{closure}} = 0$ yet $\mathcal{G} > 0$. We therefore read $\mathcal{G}$ as corroborating evidence and rest the model-dominance conclusion on the decomposition.

The sign rule, and scope. Whether the *trained* physics model outperforms the *trained* black box at sample size n follows a sign rule.

Remark 1 (Sign of the physics penalty)**.** *Write* $\mathbb{E}[\hat{R}_{\text{phys}}(n)] \approx R_{\text{e2e}} + V_{\text{phys}}(n)$ *and* $\mathbb{E}[\hat{R}_{\text{direct}}(n)] \approx R_{\text{direct},\infty} + V_{\text{direct}}(n)$*, with* R_{e2e} *and* $R_{\text{direct},\infty}$ *the two asymptotic risks,* $V(n) \downarrow 0$ *the estimation variance of each estimator at budget* n*, so that the* physics penalty at budget n *is* $\mathcal{T}(n) \approx \Delta_\infty + [V_{\text{phys}}(n) - V_{\text{direct}}(n)]$

with $\Delta_\infty := R_{\mathrm{e2e}} - R_{\mathrm{direct},\infty}$ *its* asymptotic *part and* $D(n) := V_{\mathrm{direct}}(n) - V_{\mathrm{phys}}(n)$ *the variance a constrained prior saves. Then* (i) $\mathcal{T}(n) < 0$ *(physics helps) if and only if* $D(n) > \Delta_\infty$*: the variance a constrained prior saves must exceed its asymptotic bias.* (ii) *If* $\Delta_\infty > 0$ *and the variance gap vanishes (*$D(n) \to 0$*), there is a critical budget* $n^\star$ *beyond which* $\mathcal{T}(n) > 0$ *for all* $n > n^\star$*.*

Both clauses are rearrangements of the definitions above, which is why we state this as a remark: given the two expansions, *(i)* is an identity and *(ii)* its limit. It supplies vocabulary—a name for the asymptotic part, a name for what a constrained prior saves, and the budget at which they change places—and no prediction that any measurement below is tested against. The limiting refinement in which $\Delta_\infty = B_{\mathrm{closure}} - \mathcal{G}$ when the control is asymptotically Bayes goes beyond a rearrangement, and is in §S2 with the working of both clauses. The consequence used below is that a prior is predicted to hurt precisely in the mature-data, low-noise, well-covered regime, and every physics-versus-control difference here is a penalty at the budget actually trained. We prove no "optimal grounding" theorem, and the decomposition leaves out-of-distribution accuracy where it was. Its variance term $D(n)$ carries the semiparametric phenomenon in which an estimated propensity outperforms the known-true one.[31,32] A fitted nuisance can be the more *efficient* estimator even when its bias is as large, so a true-valued input can miss the lower total risk at finite n.

2.2 Data curation: sources, splits, and labels

Solubility measurements come from BigSolDB 2.0:[33] 108,287 rows over 1525 solutes and 212 solvents. Melting points merged in as single-component crystal labels bring the modelled corpus to 127,930 rows over 20,766 solutes and 228 solvents between 243 and 438 K. We partition by Bemis–Murcko scaffold (a molecule's ring systems together with the linkers joining them, side chains removed) into 111,724 training, 8,103 validation and 8,103 test rows, so that every test solute is unseen at training. A melting-point-independent miscibility/structure filter sets row membership, and we merge on crystal T_m, ΔH_{fus}, Hansen solubility parameters and infinite-dilution activity coefficients $\ln \gamma_2^\infty$ as per-solute labels where available. The rows of a split carry different labels: of the 8,103 test rows, 5608 carry a solubility measurement while the other 2495 carry a melting point alone, so every accuracy number below is on 5608 rows and says so (§2.4). 796 of those 5608 rows, 14.2% of them, carried by 26 of the 147 test solutes, have a solute of more than one component (salts, two ionic liquids and four hydrates). For those, the solid in equilibrium with the saturated solution differs from the one-component crystal Eq. 1's ideal term is written for, and all 26 of those solutes lack a measured melting point. They stay in: dropping the stratum moves the control's mean and the grounded arm's the same way and widens the physics penalty this paper reports, so removing it at this stage would flatter the comparison (§S3.2). Solid form is uncontrolled on both label sources, the solubility rows aggregating literature measurements on whatever form each laboratory had, and the $T_m/\Delta H_{\mathrm{fus}}$ labels being merged from a separate compilation free to refer to another form. A polymorph difference therefore enters the ideal term as a systematic and the inter-laboratory disagreement below as part of its spread, and both documents leave it unseparated. Crystal supervision is scarce: the test and validation splits contain essentially no measured ΔH_{fus}, the quantitative basis of the identifiability argument (§2.1). The external single-component data we use for grounding are an open crystal-property pool (∼15,000 melting points), the VT-2005 σ-profile database, and ThermoML infinite-dilution activity coefficients.[34] Two hazards the pipeline handles explicitly (melting-point unit detection, and the instability of the seeded scaffold split across pipeline versions) are recorded in §S8.

The label-noise floor, and the two functionals it is read with. Independent determinations of the same system disagree, and that disagreement floors what any predictor can achieve. On the raw BigSolDB rows, we group by (solute, solvent, temperature to 1 K) and keep the groups reporting at least two distinct sources. Within a group, the disagreement statistic is the mean absolute deviation of $\ln x_2$ from the group mean. Aggregating that per-group statistic over groups requires a choice of functional, and the two standard ones differ by a factor of five, so we report both. The mean over groups is 0.15 $\ln x_2$ units for the 3948 groups backed by two sources and 0.31 for the 349 backed by three. The median over the same groups is 0.03 and 0.09. Collapsing exact-duplicate values first (the same measurement re-tabulated across compilations, which would otherwise deflate the floor) leaves the mean at 0.16 over 4261 groups. The scaffold-split MAE reported here (1.70 to 1.85) is above the larger functional by a factor of five, so the accuracy ceiling stands clear of label noise under either reading. This floor is a mean absolute deviation in $\ln x_2$. The aleatoric estimate published for this task, an inter-laboratory root-mean-square error (RMSE) of 0.75 in $\log_{10} S$ on 34 duplicate solutions,[1] is a different functional on a different base, and we keep the two apart below, pooling and converting neither into the other. A benchmark on this corpus argues that the inter-laboratory figure describes worst-case rather than expected disagreement and puts the floor near 0.106 in $\log_{10} S$;[35] on that reading the ratio above is the conservative one.

Disclosure: the scaffold guarantee is not certified for these runs. The released stream builder excludes any pool molecule whose scaffold appears in validation or test. One σ-stream build on disk was nevertheless generated against the wrong split files, and per-run stream provenance went unlogged, so which build fed which arm lies beyond the artifacts' reach and the guarantee is a property of the builder rather than of these runs. Of the 147 solutes spanned by the 5608 labelled test rows, 7 carry their own reference profile in that build, and 8, those 7 among them, carry a scaffold relative. The second count is therefore the union, and deleting it deletes both. The ungrounded arm carries no σ stream either, so the supervision contrast is the one a leak can reach (§3.1). The file evidence, the contrast each fact bounds, and the one molecule the leak touches in the n=44 analysis of §3.3 are in §S8.

The leak-free re-run. We fixed its protocol before its numbers exist (§S8). The ungrounded and the grounded arm run at five seeds (42 through 46) on the published COSMO-SAC configuration unchanged, on a σ stream rebuilt against *this* split's files and certified inside each training run. Both are read on the same 5608 labelled test rows and reported seed by seed. We fixed the decision it settles in the same pass, stating it in three branches rather than in the favourable one alone (gains of one sign, gains straddling zero, gains of the opposite sign), together with the list of displays the five-seed values replace. The supervision contrast is what that decision hangs on. All three branches leave the substitution result standing: it is one checkpoint evaluated two ways, beyond the reach of any σ-stream build.

2.3 Model, closure, and training

What follows specifies the instrument rather than proposing a method. None of the architecture, the closures or the training schedule is offered as a contribution, and no result in this paper turns on a choice among them; they are described in the detail they are because a measurement of a fixed closure's error is only as reproducible as the pipeline it was taken in. A shared message-passing encoder embeds solute and solvent, a bipartite cross-attention block lets them interact, and the resulting pair representation feeds every head (§S1). We implemented two further backbones and both stay outside the numbers reported here; the physics-channelled one was run against this encoder and trailed it, an observation recorded with no set, seed count or interval attached (§S1), so we claim no accuracy comparison between backbones. A fusion head predicts T_m and ΔH_{fus}, giving the ideal term $\Phi(T)$ of Eq. (1). The activity term $\ln \gamma_2$ comes from one of three differentiable closures. Two are controls: the fitted NRTL (non-random two-liquid) excess-Gibbs-energy model, and its symmetric one-parameter collapse. The third is the closure under study and is parameter-free: a σ-profile head emitting a 51-bin screening-charge-density profile per molecule, and a COSMO-SAC layer mapping the profile pair to $\ln \gamma_2$.

External single-component data enter through *grounding streams* (the training-time sense of grounding). Each stream is a sidecar of *self-solvent* rows (no solubility label, only the target factor's mask active) interleaved into training under the guard above. It grounds the crystal factor from the melting-point pool and the activity factor from the σ-profile database, and so identifies from external data a factor that solubility alone leaves non-identifiable and that a black-box predictor is unable to use. Training runs a three-stage curriculum under a weighted loss, and the stages are named by number wherever one of them is referred to below: *P1*, a warm-up on the auxiliary streams alone; *P2*, joint training against solubility; *P3*, a low-rate stabilisation stage. The schedule pinned for the arms of Fig. 2 freezes the σ head at the end of P1, so the σ stream enters the warm-up only.

The in-house closures, and where their validation stops. The dispersion-off closures in the fidelity measurement (§3.5.3) are our own differentiable implementations, so those arms share one code path, and each consumes its own model-prescribed σ-averaging (Mullins for 2002, Hsieh for 2010). Our 2002 layer (the *2002, all channels* arm wherever the kernels are compared, so named because it sums the segment contributions of all three profile channels rather than the nonpolar one alone) reproduces the NIST reference COSMO-SAC-2002[36] to RMSE 0.0035 $\ln \gamma$ on identical VT-2005 profiles. Our COSMO-SAC-2010/dsp layer,[37,38] built on the *typed* latent of three donor/acceptor-resolved σ-profiles per molecule in place of one, reproduces the NIST 2010 reference with dispersion *off* to RMSE 1.6×10^{-4} $\ln \gamma$ over 400 pairs, the largest single departure being 1.7×10^{-3} (§S1). The dispersion term is *not* validated to that standard, so we compute every dispersion-off contrast on our layers and quote the *2010, dispersion on* runs from `cCOSMO`, that reference implementation itself.

The same closure against a published evaluation. Reproducing a reference implementation fixes the code, and it leaves open whether the error this paper decomposes is the error a published COSMO-SAC has or something peculiar to this data assembly. Table S27 answers that: scored over the same records as a published evaluation of the same closure,[39] our differentiable layer lands inside the bracket that evaluation spans across profile databases, under both conventions. Two limits stay open: the profile database is the one axis we cannot match, and whether its 6977 points are the rows of our 7889 is unproven. Subject to them, the closure error decomposed in §3.5 is, in average absolute deviation (AAD), the error a published COSMO-SAC has. That evaluation also scores a donor/acceptor-typed later kernel on those records at about half the deployed closure's AAD. Every ceiling this paper reports is therefore this kernel's ceiling and not COSMO-SAC's, and the results below are substitutions inside a fixed map and not claims that the map is the best available.

2.4 Evaluation protocol, and the DirectGNN control

In the *oracle substitution* we replace the predicted σ-profile $\hat{z}$ with the reference VT-2005 profile $z^\star$ before the closure, matching by canonical SMILES. One object is replaced, not two. The σ-head emits a normalised 51-bin shape and a cavity area whose product is the area-weighted profile the closure reads, and the tabulated pair stands in the same relation. The reference area therefore reaches the residual term's A_2/a_{eff} prefactor as part of that single replacement (§S1).

A property of our head bears on what that substitution does, and it is a defect rather than a choice. The encoder carries a separate learned adapter for the solute and the solvent role, and the σ-head reads whichever the molecule occupies, so one molecule has *two* predicted profiles. A σ-profile is a property of a molecule alone, and the closure reads the pair, so the model can and does violate an identity no activity model may violate. On a self-pair, $\ln \gamma_2$ must be exactly zero. We fed every one of the 2634 distinct test solutes to both roles at its own temperature. Of those, 102 are untestable, the solver sitting at $x_2 = 1$ there, where the residual term vanishes identically, so $\ln \gamma_2$ is zero by composition rather than by profile agreement. On the 2532 that are testable, the median $|\ln \gamma_2|$ is 0.154, the mean 0.286 and the largest 5.78, with 95% above 0.01. With one profile on both sides, the same computation returns zero to machine precision, so the role gap is the whole of the violation. The size of the repair depends on which of the readout's three numbers carries that gap, and on these arms it is the profile. With the two roles agreeing on the shape the residual bracket is identically zero whatever the areas, so on a self-pair the profile disagreement is the whole of the violation. §S1 prices a mismatch on the area and on the volume, and gives how far the deployed arms' areas sit from the tabulated ones. The substitution overwrites both roles at once and therefore closes that gap as well, which would be a second thing it changes besides the profile values. On the rows we score it changes nothing, and the reason is structural rather than empirical. The residual term takes the solute's pure-component reference from the solute-role profile itself, so on a pair of distinct molecules each is read in exactly one role and no role discrepancy enters the bracket; only a self-pair puts one molecule in both positions. No scored row is one. Counted on canonical SMILES, the labelled and unlabelled test rows, the validation split and both activity sets contain no self-pair at all, and the single instance anywhere in the corpus is a water-in-water row of an auxiliary stream, which carries no solubility label and is never scored. The substitution therefore changes the profile values alone on every row this paper reports, and tying the roles remains the repair for the identity itself, which is a retraining. We match solute and solvent independently. The substitution replaces those two profiles and stops there, the crystal head reading a solute-only embedding it leaves untouched. The bounded correction does read the closure's own parameters, so the crystal values the solver finally receives shift by a little: pooled over the five seeds, at most 3.2×10^{-5} in Φ against a median $|\Phi|$ of 4.5, on the 8066 of the 8103 test rows where the substitution applied and on the 5608 labelled rows alike (§S1). The reference table covers 99.3% of the labelled test rows on the solvent side and 5.3% on the solute side, so the solvent profile carries the substitution contrast below.

We apply it three ways to rule out implementation artifacts, differing in *when* the replacement is made: at evaluation alone; injected at training so the model co-adapts; and a channel swap. The last is injected at training under a coordinate-descent schedule that freezes the crystal branch during the SLE phase, so the activity branch alone refits against it. The last two are at seed 42. An analogous override substitutes reference crystal targets. The activity target for the decomposition is the infinite-dilution $\ln \gamma_2^\infty = \lim_{x_2 \to 0} \ln \gamma_2$, on two sets that differ in their reference database and their regime (§3.5.1–§3.5.2).

DirectGNN shares the encoder, interaction, readout and pair representation, adds a thermometer (cumulative-threshold) temperature encoding, and maps directly to $\ln x_2$ with no heads, closure or solver, so it sees T explicitly where the physics arm sees it through $\Phi(T)$ and the closure. We tuned the optimisation settings per arm, and they differ by up to an order of magnitude. The TGNNSolv−DirectGNN gap is therefore a comparison of two separately tuned pipelines that differ in the thermodynamic bottleneck, and not an isolation of it; every physics-penalty comparison below carries that caveat (§3.7).

We report the controlled comparisons of §3.1 as mean±sd on a fixed row set, at five seeds for the arms the leak-free re-run trains and three for the rest. That lock is the 5608 test rows carrying a solubility measurement, 5608 of the 8103 (§2.2), so all arms score identical rows. The 2495 rows outside it carry a melting point alone and are unscorable in every arm alike, the control included, each dropped for its label and not for anything an arm predicted. Which rows carry a label fixes the set: the lock's finite-prediction clause is inert, removing no row from any arm at any seed, and §S3.2 gives the decomposition. The runs behind every number reported here fall into twelve families, a family being a set of runs sharing a script, a corpus and a seed policy, and Table S29 gives each number its family, its seed count and the artifact it regenerates from.

2.5 Estimating the split: one cell, one convention, one rule

The instrument is a procedure taking three inputs and no others: a fixed closure g, a table of reference values $z^\star$ its argument can be read at, and measurements m of its target on rows where both exist. Per chemically defined row set it returns five things: the closure's own error MSE, an upper bound on what the inputs leave unresolved, the margin between them, a cluster interval on that margin, and a verdict of admissible or otherwise. The verdict carries the test that decided it. Four analyst choices set the binning bound of Lemma 1(c): the bin count, where the equal-count bin edges fall, the within-bin variance convention, and the combinatorial convention. We fix all four here, before any margin is reported — a data-set contract in the sense Nael et al.[40] argue for, declaring the processing rules, the end point and the split before the claim rather than after it — and one setting of them, with the row or the molecule pair as unit, is an *estimator cell*. The map of §3.5.1 is what running the procedure on this closure returns. Lemma 1 supplies two bounds and we declare here which one every margin is read from. The constant-offset bound of part (b) is strictly slacker than $B_{\text{closure}} \geq \text{MSE} - B^{\text{up}}_{\text{insuff}}$, which part (c) already gives, under both combinatorial conventions and on both sets, so it adds nothing to the ordering and is reported for completeness alone (§S1.1). Every margin in this paper is the binning bound's.

The cell, and what was fixed before any margin was seen. The cell is eight equal-count bins of $g(z^\star)$, the *unbiased* (Bessel-corrected) within-bin variance, the row unit and the *residual-only* combinatorial convention. It is applied identically to every set and every stratum whatever its size, and every margin in this paper is read at it. The bin count is the choice with the most leverage: it takes the broad set's aggregate margin from +0.01 to +0.96. The cell therefore never stands alone where a finding rests on it, and we run the same sweep for the strata the rule passes and report it with them. One convention governs both sets rather than the favourable one on each, which costs the broad set's margin about half its value.

Admissibility then asks two things of a stratum and a third of the cells. *(a)* The stratum must be *boundable* at that cell: $n \geq 40$, eight bins holding five rows. *(b)* Its margin must keep its sign when each contributing source publication is deleted in turn, *in all four unit × convention cells*. That clause is load-bearing: dropping it lets the set taken whole pass. *(c)* Admissible cells holding identical rows count once, a row set being demoted to a larger one only when the residual between them falls short of admissible. Clause (c) is rebuilt on every draw of the multiplicity null, on the permuted labels, like the rest of the rule. The stratification reads a SMILES string and stops there, leaving m, g, the squared error, the source and the bootstrap unread, so every cell was fixed before its outcome — a property of the code and not an assurance. §S1.1 states all four rules in full, with the class cascade, the demotion test and what each of them costs.

3 Results and discussion

Every claim below carries its set, its size and its scope in the sentence that makes it, and §S3.1 collects the same nineteen side by side as one table (Table S3). Two reading conventions travel with them. A cluster-bootstrap frequency P is the frequency, over resamples, with which the margin it is attached to stays positive — a resampling frequency and *not* a p-value against any null. And every interval below is a percentile interval over the unit named beside it. §S1.2 gives those units, the draw counts, the rounding rule the counts support, and the two spans that are of a different kind.

3.1 Supervising the intermediate toward the reference, and substituting it at prediction time

The word "grounding" names two opposite operations on the same VT-2005 database. Every solubility figure in this subsection and the next is an $\ln x_2$ MAE over one row set, the 5608 of the 8103 scaffold-split test rows that carry a solubility measurement (Table S1). The single-seed control arms below carry that identical set, so every comparison here stays inside one row set. Supervising the σ head against the database *during training* moves solubility MAE from 2.037±0.064 (ungrounded) to 1.927±0.094 (grounded) over five leak-free seeds. The mean of +0.11 is not a sign. The per-seed gains are +0.149, +0.122, +0.175, −0.017 and +0.122, the two arms' per-seed values interleave, and at seed 45 the grounded arm is the worse of the two. The two arms differ in training budget. The grounded arm precedes the shared curriculum with a σ warm-up on the grounding pool of up to 40 epochs, early-stopped on a held-out tenth; the ungrounded arm sets that budget to zero (§S3.2). We left unrun the warm-up against permuted σ targets that would separate the two, so the gain prices the intervention entire, extra pretraining and accurate targets together, rather than the accuracy of the reference values by itself. Those spreads are over initialisation and batch ordering, so they price the training run rather than the draw of test solutes. Pooling the five seeds and resampling the 147 test solute clusters, we put the gain at +0.110, 95% percentile interval $[+0.001, +0.222]$. The two prices hold different things fixed and they do not agree: over the draw of solutes the interval clears zero at its endpoint, and over training runs the gain has no sign at all. The ungrounded arm carries no σ stream, so this is the paper's only contrast a σ-stream leak could inflate, and it is what the re-run was pre-committed to settle. Its stream is certified inside each training run and reaches no test molecule: 0 by canonical SMILES against 5101 held-out SMILES, and 0 by Bemis–Murcko scaffold against 2702 held-out scaffolds (§S8). On these arms there is nothing to delete. The published three-seed family, whose stream build is uncertified, put the same gain at +0.198 $[+0.06, +0.33]$, and at +0.172 $[+0.03, +0.31]$ with the eight leak-reachable solutes deleted, on 351 of the 5608 rows. The published family's two values bound where a leak can act *there*, and are what limitation (xii) refers to. The leak-free build supersedes them: in it a leak cannot act at all. Substituting the reference profile for the learned one *at inference* degrades it, and that substitution alone is the effect this paper is named for. It erases any positive R^2 (Fig. 2): the model predicts solubility better *from its own learned σ-profile* than when *given the external reference one*, at MAE 1.93 against 2.34 in $\ln x_2$ units. The gap holds over the same five seeds, and the two arms' per-seed values are again disjoint. Resampling the same 147 solute clusters over the pooled five seeds, we put that penalty at +0.408, 95% percentile interval $[+0.29, +0.54]$. The median per-row gap is +0.30 against the mean's +0.41, so the effect sits at the middle of the distribution and grows heavier above it. Of the figure's eight arms, two are non-COSMO controls, DirectGNN and a fitted NRTL closure. Five COSMO-SAC arms differ in how the σ head is trained (the warm-up asymmetry above included) and in what it is fed. The sixth (*grounded+comb.*) differs instead in the closure, restoring the Staverman–Guggenheim combinatorial term. *Ref. σ (eval)* is the grounded checkpoint re-evaluated with the reference profile substituted at test time, and *ref. σ (train)* the co-adaptation control injecting it during training instead.

The +0.41 penalty is the evaluation-only substitution, and part of it may be unrelated to the closure: swapping an input distribution at test time degrades a pipeline on its own. Separating that component needs the reference profile injected *at training*, so that the model co-adapts to it, and no such arm was run at five seeds. The share of the substitution gap that co-adaptation would carry is therefore unmeasured here, and the penalty below is the two operations together (§S3). The five-seed arms are the ones that can be set against one another, and there the evaluation-only substitution costs several times what supervision gains ($+0.41$ against -0.11). Both carry the co-adaptation confound, so that ratio measures something other than a co-adaptation-free magnitude, and this work supplies no arm that does. The -0.11 has no sign of its own either: its per-seed values are $+0.149$, $+0.122$, $+0.175$, -0.017 and $+0.122$. The ordering of the two operations is therefore unresolved, and only their signs are carried forward.

3.2 What the substitution costs: level, spread, and the choice of solvent

Figure 3 sets the same rows out predicted against measured, for the control and three of the closure arms. Two things are visible there that the arm ordering hides. First, every arm is severely shrunk toward the corpus mean. The ordinary least-squares slope of predicted on measured is 0.515 for DirectGNN, 0.399 ungrounded, 0.511 grounded and 0.658 with the reference profile, all of them well short of 1. In every panel the median trace crosses the identity: sparingly soluble compounds are called more soluble than they are and highly soluble ones less. All four slopes are seed 42's, the figure's seed, with one run behind each. The shrinkage originates outside the closure: at that seed the closure-free control's slope differs from the grounded closure's by 0.004, so whatever produces it lives in the encoder and the data. Second, the reference-profile arm is the *least* shrunk and the worst. Its slope is the closest of the four to 1 and its prediction spread exceeds the measurement's, yet its MAE is the highest and its R^2 is indistinguishable from zero. Substituting the reference restores spread that is misaligned with the truth, and its worst region is the sparingly-soluble tail it appears to reach. The closure is driven hardest in that tail. At seed 42 the substituted arm returns 119 rows with $\ln \gamma_2 > 10$ where the learned arm returns none, the learned arm's largest being 7.77 against the substituted arm's 14.6. And 141 of its $\ln x_2$ predictions fall below -15, to a minimum of -22.3 where the corpus itself stops at -17.7. The worst 500 of the 5608 rows carry 0.25 of that seed's $+0.26$ mean gap. We score both arms at the 30 segment iterations of §2.3, with the fixed point left short of convergence there. Those rows are therefore the ones on which the truncation is least certified, as well as the ones on which the two profiles differ most. The retained artifacts keep none of these arms' weights, so what the count is worth on them cannot be measured here. The tree does keep one trained COSMO-SAC head (a retired run whose weights were fit at 8 iterations, and which scores 2.61 at 30 against 1.92 at 8 on these same labelled rows). On it the substitution penalty moves with the count as well. Substituting the reference profile costs $+4.32$ $[+3.61, +5.25]$ at 8 against $+6.43$ $[+5.39, +7.68]$ at 30, over 5571 rows in 64 solvent clusters whose solvent carries a reference profile. That head's penalty departs from the one reported here in both size and direction: fifteen times larger at the count the $+0.41$ is measured at, and moving its predictions the other way. Those two figures price the lever, not this paper's $+0.41$. The restored, misaligned spread is the misspecified map made visible on the solubility axis.

The substitution also costs the decision the model is used for. The separate-tuning confound of §3.7 leaves this comparison clear: the reference-profile arm re-runs the grounded checkpoint's own weights over the same rows and changes exactly one thing, the profile fed to the closure. One tuning serves both sides. They differ in Φ by at most 3.2×10^{-5} over these 5608 rows at the worst of the five seeds. Co-adaptation is the confound that survives. Like the MAE gap above, this is an input swapped at evaluation; we ranked no arm that had trained on the reference profile, and what co-adaptation would do to an ordering stays unmeasured here. We group the labelled test rows by solute and temperature and rank the solvents inside each

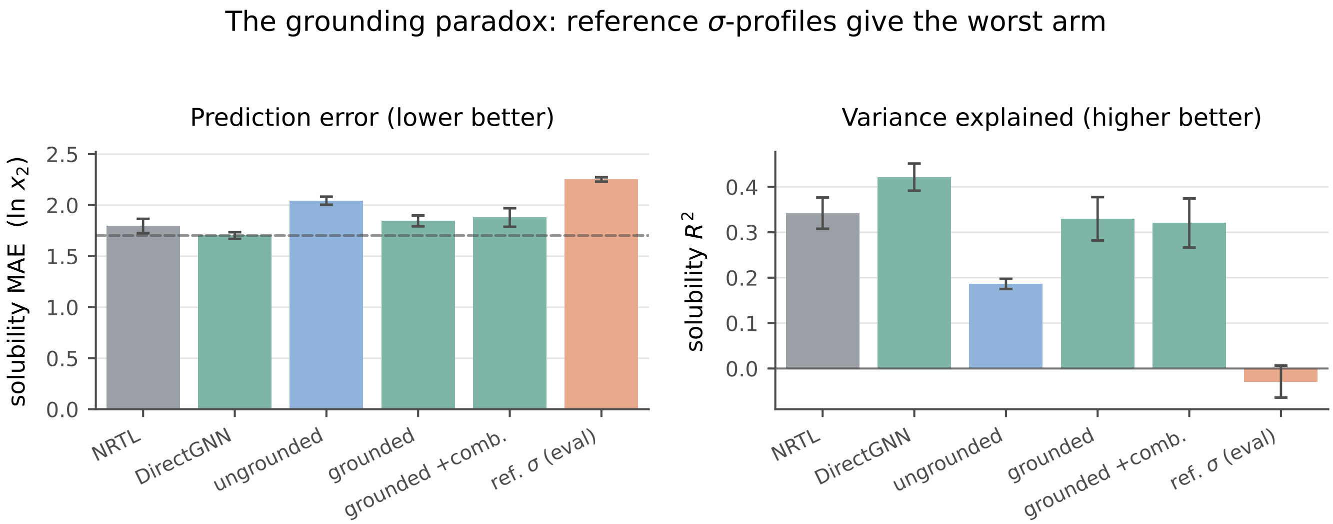


Figure 2: Controlled comparison on the 5608 labelled test rows of the scaffold split. Bars are five seeds, mean ± a population standard deviation over them. The ungrounded, grounded and σ-oracle arms come from the five-seed leak-free re-run. Dashed in the left panel alone, the DirectGNN reference.

group (823 groups over 107 solutes; Table S22). The mean per-solute Spearman correlation then falls, pooled over the five seeds, by -0.185 $[-0.27, -0.11]$, every per-seed interval excluding zero. A rank correlation is defined for both arms in 819 of those groups, and Table S22 gives the arms seed by seed. Four of the five falls exceed any difference between seed replicates of a single arm on this measure, the largest of which is 0.132. The fifth, at seed 42, is 0.114 and does not: at that seed the fall sits inside the spread one arm shows between its own seeds. The deposit carries that benchmark for the rank correlation alone, so we read it against that measure and leave the two below unbenchmarked. Every interval here is a 95% percentile bootstrap over those 107 solute clusters, the effective sample rather than the 823 groups. Two decimals is the precision the draw count determines, coarser than the precision the deposit records. At the 4000 draws behind them a third decimal is a property of the bootstrap seed, so we apply to these the coarsening rule §S3.8 states for the map. Normalised discounted cumulative gain over the top three solvents (nDCG@3) falls by -0.071 $[-0.11, -0.03]$, and top-1 accuracy by -0.088 $[-0.17, -0.01]$. We quote that last figure pooled, and pooled only: three of its five per-seed intervals span zero, so of the three ranking metrics it is the one whose sign the per-seed columns leave open. In 42 to 53% of groups the model selects a different solvent, and asymmetrically. The learned profile by itself picks the measured best solvent in 18 to 24% of groups, the reference profile by itself in 10 to 15%. All of that is an ordering rather than a level. A separate affine recalibration fitted inside every group removes 91% of the MAE gap, $+0.426$ $[+0.30, +0.55]$ to $+0.040$ $[-0.00, +0.08]$. The map carries a slope as well as an intercept, the largest concession a miscalibration reading can ask for. What is left of the gap no longer stands clear of zero. A per-group map that absorbs it is the restored, misaligned spread of the paragraph above in another currency. The map is fitted on the rows it is then scored on, two free parameters against a median of five solvents, so it bounds what the concession can buy rather than measuring it. Refitted leaving out the row it scores, it returns $+0.32$ $[-0.61, +1.20]$ pooled, an interval wide enough to hold both the whole gap and none of it. On the 550 groups that hold at least five solvents, where a refit has something to work with, it runs $+0.14$ to $+0.28$ by seed with three of the five intervals spanning zero (§S6.1). The MAE penalty is level and scale in the sense that a per-group map fitted on the answer absorbs most of it. The interval is wide for a reason the design fixes rather than the answer: a two-parameter map fitted on the other two rows of a three-row group is exactly determined. Excluding only those groups, 87% of the cells remain and the out-of-sample gap is $+0.40$ $[+0.13, +0.78]$; at floors of five and six solvents it is $+0.20$ $[+0.04, +0.37]$ and $+0.17$ $[+0.01, +0.35]$, excluding zero at each. The recalibration that absorbs 91% in sample therefore absorbs at most about half of the gap out of it, and what remains is positive wherever the design can resolve anything at all. The raw gap falls along the same sweep, from $+0.43$ to $+0.29$, so these floors are not nested samples of one population and part of that decline is chemistry rather than recalibration. The ranking penalty survives both fits, by invariance rather than by any centring argument. The invariance is to an *increasing* affine map. The fitted map decreases in 10 to 15% of learned-profile groups and 15 to 21% of reference-profile ones, and there it reverses that group's order outright. Restricted to the 485 groups of 823 whose map increases in both arms at all five seeds, the ranking gap is ρ -0.107 $[-0.179, -0.031]$, with nDCG@3 and top-1 then spanning zero. Only the rank correlation survives that restriction, and the subset it survives on is 59% of the groups.

The ordering that stands in its place is indistinguishable here from a solute-blind one. Two references are blind to the solute: the same arm's own predictions for a randomly drawn other solute over the same solvents, and a ranking of those solvents by their leave-one-solute-out mean measured solubility. Set against both, the learned-profile arm is above them on Spearman, Kendall, top-1 and nDCG@3 at every seed: all forty of its margins are positive, and 27 of the forty intervals exclude zero. The thirteen that do not are concentrated rather than scattered, seven of them at seed 42 and four at seed 45. The separation is therefore a property of three of the five runs, not of every one. The reference-profile arm's own forty margins are smaller, 36 of them positive. Their intervals carry the reading: 36 of the forty include zero, and the four that do not are all one reference's, the donor floor at seeds 43 and 44 (Table S22). The intervals measure a failure to separate the reference-profile arm from solute-blind orderings on most of the design, and they stop short of demonstrating that the two are inseparable. Both arms are above chance, the within-group permutation floor being $\rho = 0.00 \pm 0.02$ for each. Supplying the true physical intermediate costs an ordering rather than a calibration: the arm's ordering of solvents becomes indistinguishable from one drawn blind to the solute.

3.3 What moved inside the model, and whether a downstream correction repairs it

A candidate *mechanism* for the low-fidelity, end-to-end regime: a σ-profile head trained *end-to-end* through a fixed, misspecified closure, with no σ supervision in that phase, is under pressure to *cancel* the closure's error instead of being physically faithful. This section establishes the first of its two parts. The antecedent, that the latent leaves the reference along a direction shared across molecules, is measured. The consequent, that the departure cancels part of B_{closure}, stays unmeasured, and the one direct check argues against it. A second candidate for the same drift is live and is not separated from this one: the deployed ideal term sets ΔC_p to zero, which is one-signed and worth a median 0.32 in $\ln x_2$ where a melting point is measured, so a polar drift is also what compensating an over-large Φ would look like. That alternative is generated by a simplification of ours rather than by the chemistry, and nothing here tells the two apart (§S5). We compare the predicted profile $\hat{\sigma}$ against the reference VT-2005 profile molecule by molecule (n=44): anatomy of the object the two operations of §3.1 act on, and not evidence for the sign of either.

What is measured, and on what. We compare within a single grounded model, checkpointed before and after solubility fine-tuning, over the n=44 matched molecules, at three seeds. Those runs are a *separate* set from the arms that produce the paradox: their σ head

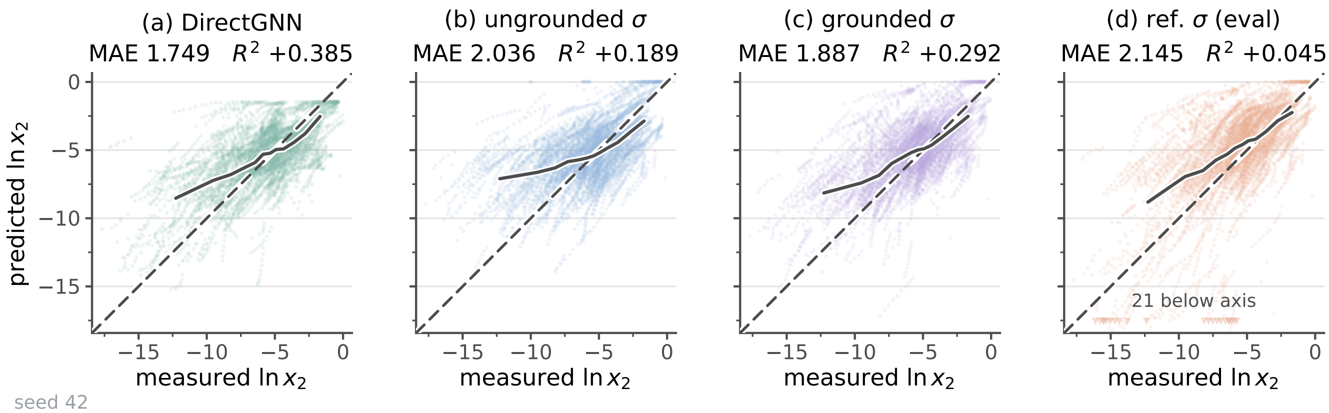


Figure 3: Solubility parity for four arms of the controlled comparison: predicted against measured $\ln x_2$ on the 5608 labelled test rows of the scaffold split, which are 763 solute–solvent pairs over 147 solutes and 70 solvents measured between 248 and 368 K. (a) DirectGNN, the closure-free control; (b) COSMO-SAC over an unsupervised σ-profile; (c) the same closure with σ-profile supervision during training; (d) the arm of (c) with the reference VT-2005 profile substituted at evaluation. Dashed, the identity. Solid, the median prediction in twelve equal-count bins of the measurement. Each panel header carries that panel's own MAE and R^2. Both axes span the measured range with a margin around it, and the markers along the foot of (d) are its 21 predictions that fall below the frame, each at its own measured value. Panels (b), (c) and (d) come from the five-seed leak-free re-run: the ungrounded, grounded and σ-oracle arms respectively.

stays unfrozen through fine-tuning, where the arms of §3.1 freeze it from P2 onward. The deposit cannot recover those arms' own drift. What follows is therefore measured on runs of its own and stays unestablished of the checkpoints whose paradox it is offered to explain. Under σ supervision the head reproduces the reference to within $36 \pm 1\%$, so the intermediate *can* carry the physical shape — as an in-sample fit, all 44 probe molecules lying inside the σ-grounding pool. Fine-tuning drives it a further $41 \pm 3\%$ away, a total departure of $51 \pm 2\%$ whose magnitude is unstable across analysis configurations. These runs therefore fix the sign and a direction shared across molecules, not the size. On the reference ladder (§S5) it sits at 0.507, past the 0.493 of one corpus-mean profile reused for every molecule: the drift consumes essentially all the specificity grounding supplied.

What the working shows, and where it is. Whether the drift is structured beyond a constant offset rests on the mean-centred spectrum and the three nulls it is read against, of which it clears one. Those, the distortion operator and the ratio at which it transfers, and the arm that freezes the head at its supervised warm-up state are in §S5. So is the closure's Fisher geometry, effectively rank one at infinite dilution, which is what leaves σ free to move at little cost to the fit. Whether the departure *cancels* closure error remains unshown. The one direct check is under-powered and runs against it. That check and the closure's own-axis sweep both sit on the infinite-dilution axis, which is where cancellation could have been measured, so the reading stays inferred from the paradox itself. What is established is weaker and still consequential: the intermediate departs from the molecule's charge distribution, the risk misspecification poses to physics-grounded latents[30,41] and the collapse an adapted representation can suffer.[42] Its low-rank *structure* excludes a random collapse. Two accounts that would displace this one are answered where they arise: TeNNet-SAC in §1 and the ill-conditioning reading in §3.4, both with their working in §S11.1. §S5 is the gated output correction that fails to recover the accuracy.

The learned profile is chemically inadmissible in the window the closure's

hydrogen-bond term reads. COSMO-SAC's hydrogen-bonding contribution acts on segments beyond $|\sigma| = \sigma_{\mathrm{hb}}$, and the 2002 kernel assigns a segment to the donor side by that threshold alone, with no atom typing. Donor-window area is therefore a statement about surface charge density and not a hydrogen-bond-donor count. Of the 1003 molecules in the reference tabulation that lack an N–H, O–H or S–H, 264 have area there, and acetonitrile's 6.71 Å^2 stands above ethanol's 6.20. The molecule-matched comparison carries the argument. Cyclohexane, acetone, toluene and tetrahydrofuran each carry 0.000 Å^2 of donor-window area in the reference tabulation, exactly zero and not merely small, against 15.75 for water. The learned profile carries 44.0, 32.4, 24.5 and 50.7 Å^2 in that same window, between 23% and 36% of each molecule's total surface. Figure 4 draws that comparison over the sixteen solvents carrying the most scored rows; the reference is exactly zero in twelve of them. We read those areas off the one trained COSMO-SAC head the tree holds, the retired run of §S5's two-MSE check, scoring MAE 2.61 at $R^2 = -0.31$ on the scaffold split. A null arm in the same deposit separates architecture from training. As a fraction of profile mass in the same extreme bins, where the reference tabulation's median is 0.000, an *untrained* network of this architecture reaches 0.212 against the trained head's 0.189. The window is full before any training, and training emptied it slightly.

Two causes stay entangled. This head's σ-supervision stream ran for zero steps, and §S6.2 is why that matters. Solubility alone underdetermines the profile's shape, leaving an unsupervised profile free in exactly the bins that carry no constraint, which accounts for a quantity of this size. The parameterisation accounts for far less. The profile is emitted through a softmax over the 51 bins, strictly positive where 57% of the reference table's bins are exactly zero, so supervision at any strength leaves the window occupied. But a softmax reaches 10^{-6} at a logit gap of ten, four orders below what is measured. The measurement establishes that the window is occupied in a trained arm, and that the closure's hydrogen-bond term is consequently live on every pair of the scored set, a defect whichever cause carries it. A supervised arm separates the two, and a head able to emit zeros removes the architectural half.

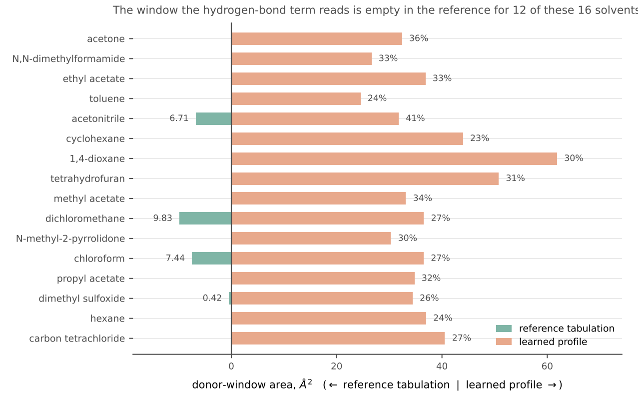


Figure 4: **The window the hydrogen-bond term reads, in the reference tabulation and in the learned profile.** The sixteen solvents carrying the most scored rows, ordered by that count. Left of the axis, the reference tabulation's donor-window area, which is exactly 0.000 Å^2 in twelve of them and is drawn with its value where it is not; right of the axis, the learned profile's area in the same window, annotated with the fraction of that molecule's total surface. Areas are read off the one trained COSMO-SAC head the tree holds—a retired run fit at 8 segment iterations and scored at 30, at MAE 2.61 and $R^2 = -0.31$ on the scaffold split. They price the quantity, not the arms of Fig. 2.

3.4 What the substitution does not decide, and what would settle it

We do not resolve which of the two channels the substitution fails through. Drug-like solutes are essentially absent from the VT-2005 set, so the substitution falls almost entirely on the *solvent* channel, and the both-channel substitution the attribution would need covers ≈7 solutes. Enlarging that set means computing new reference profiles rather than looking them up.

Why it hurts stays open too, and for a stronger reason than a test that came back empty. The reading it invites is that handing the closure a faithful input surfaces a misspecified map instead of repairing it, so the reference profile is the better input where the map that consumes it is faithful. An account from outside chemistry would reach the same place without any of that. Wu et al.[43] substitute near-exact Reynolds stresses into the Reynolds-averaged Navier–Stokes equations, recover large velocity errors, and trace them to ill-conditioning of the fixed model. It does not carry here, and the reason is the size of the step. The reference profile stands 1.53 $[1.39, 1.80]$ times the learned profile's own norm away from it, and a random direction of that same norm amplifies *more* often than the substitution does (§S11.1). The reading is about a fixed map's own error and about what its inputs fail to carry, and on the solubility axis those two terms stay fused. The solubility axis is where §3.1 measured the substitution, so both quantities lie beyond observation on those rows. §3.5 measures a second axis and leaves the first unexplained.

Two experiments would settle it and both remain to be run. The first is the transfer back to the paradox's own regime, which is limitation (i). The second is limitation (x). The fidelity contrast of §3.5.3 compares *oracle-input* activity error across kernels, so $\mathcal{G}$ itself stands measured under no modern closure on either axis, and whether the *solubility* paradox survives a 2010/dsp closure is untested. Both need a typed σ-profile head trained end to end through the 2010 kernel, and the solubility one additionally needs a UD $\cap$ BigSolDB matched set. The set exists but is thin: 437 of the 5608 labelled test rows carry a UD profile on both sides, over 11 solutes and 57 pairs, so it prices a handful of solutes rather than the corpus.

3.5 A second axis: bounding a deployed COSMO-SAC against its inputs

Our estimator is the binning bound, read at the estimator cell §2.5 fixes in advance and reserves to this section. The cell is eight equal-count bins of the conditioning variable, the residual-only convention, and the unbiased within-bin variance, with admissibility clauses (a)–(c) deciding which strata may be read at all. We fixed all four before computing any margin, and §3.1–§3.4 are measured with other instruments. The decomposition of §2.1.1 separates a fixed closure's error into a part in the map and a part in its inputs. Its output is a map over chemical strata rather than one verdict for a set, for the reason §3.5.1 establishes. It runs on the two data sets of Table S1, which we keep apart throughout.

One reading rule governs every margin below, and it is Lemma 1's. A positive margin establishes $B_{\text{closure}} > B_{\text{insuff}}$. Its size is a *lower bound* on $B_{\text{closure}} - B_{\text{insuff}}$ and not an estimate, so an interval on a margin bounds that bound. A margin at or below zero leaves the two terms merged and licenses nothing, their equality staying open. B_{closure} throughout is the error of the fixed map *read at a computed profile*, so a positive margin is a property of that composite and not of the kernel alone (§3.5.3). The one cell serves both sets, both kernels and every stratum, so every cell below was fixed before the number it returned was seen.

3.5.1 The broad IDAC set (n=477)

The closure reads a temperature as well as a profile pair, so the conditioning variable is its full argument $(z^\star, T)$: the concatenated UD profile pair with T, 103 dimensions against n=477 ($\dim/n \approx 0.22$). The target m is the experimental $\ln\gamma_2^\infty$, on one data set and one profile database, UD throughout. Its chemistry is small. Every solute has 12 heavy atoms or fewer (median 7), against a median of 14 and a maximum of 122 for our own drug-like solids, and 0.46% of corpus rows carry a solute from it. The word *broad* therefore names the reach of the γ^∞ corpus the kernel is calibrated on: it covers the solvent side of deployment and leaves the solute side open. Matching to UD is by InChIKey, the hashed structure identifier, with a first-block fallback, looser than the VT-2005 rule. A displaced $z^\star$ inflates B_{closure} and the binned $B^{\text{up}}_{\text{insuff}}$ together, so its direction on the margin is unsigned. The re-estimate carries the point: the margin holds on the 465 exactly matched rows (§S8).

Every MSE, bound and margin on this axis is in squared $\ln \gamma_2^\infty$ units, a different axis from the $\ln x_2$ one the MAEs of §3.1 are measured on. The *2002, all channels* arm scores MSE $= 1.902$ against $\mathrm{Var}(m) = 4.85$. The binning bound gives $B_{\mathrm{insuff}} \lesssim 0.70$, hence $B_{\mathrm{closure}} \gtrsim 1.21$ and a margin of $+0.51$ for the set taken whole. The margin is positive in every cell of the bin-count $\times$ variance $\times$ convention grid, and two of its three bootstrap clusterings extend just below zero (§S3.3, §S3.11). The output is the map below, because the aggregate is unstable under composition. Dropping the two chemistries the error concentrates in, the glycol-ether solvents and the water-as-solute rows, takes $+0.51$ to -0.12 on the 258 remaining. The pair unit alone gives $+0.27$, and the two together -0.38 on 136 pairs ($+0.95$, $+0.28$ and $+0.01$ under the full convention). One publication carries much of it: deleting `10.1021/acs.jced.7b00114` (37 rows, six pairs, 32% of the set's squared error) takes the margin to $+0.18$. The sign survives that deletion and each of the other fourteen, and reverses only under the compounded cut (§S3.7). Every margin we report rests on the binning bound (Table S10). The corroborating estimators say that bound is slack, not that B_{insuff} is negligible (§S3).

The map, and how much of it survives the rule. We run the instrument over 59 strata of the broad IDAC set (477 rows, 185 pairs, 78 solutes, 36 solvents, 15 source publications). Fifteen have $n \geq 40$ and are boundable, which is the condition for the rule to act on a margin. Tables S16 and S17 print all fifteen, and §S3.8 the other 44, the 24 carrying no estimate at all included, each with the test it fails. Six of the fifteen are *admissible*, and clause (c) makes those six three distinct row sets, of which one is both positive and left in place by its residual test. **The glycol-ether solvents are the one row set the rule leaves standing, and we report that as the instrument's output rather than as a finding of this work.** A chemistry-blind relabelling of the same molecules reaches its margin in 10.1% of draws (below), which is the frequency at which the search would produce it on chemistry that carries no signal. Its clusters are few and its chemistry narrow, and both belong in the sentence that states it. It is 182 rows over 43 pairs, carrying 45% of the set's squared error. The solvents are 4 glycols of one homologous series, the solutes 19, every one a hydrocarbon or a halobenzene, and two of three publications carry 177 of the rows. On them the closure scores MSE $= 2.25$ against an input-insufficiency bound of $B_{\mathrm{insuff}} \lesssim 0.11$, hence $B_{\mathrm{closure}} \gtrsim 2.14$: the closure's own error exceeds what input insufficiency can account for by at least $+2.04$, 90% interval $[+1.25, +2.87]$ at the residual-only convention and $+2.94$ at the pair unit under the full one. The pre-declared cross-fitted bound, where that estimator is valid, puts the same excess at $+1.76$ $[+1.02, +2.49]$. The binning margin is positive in all four unit $\times$ convention cells at $P > 0.95$ and under every single-publication deletion, the worst leaving $+1.60$. Deleting each glycol in turn is the sharper test, three publications mapping onto four molecules, and the sign survives all four deletions, at $+1.72$ to $+2.29$. Taken one glycol at a time instead the margin spans a factor of two and a half and is set by the shortest oligomer (§S3.3). Either spread, and not the interval, is the honest width on the solvent axis, the bootstrap resampling the same four molecules the deletions delete. The margin survives the loss of the alkane solutes (74 rows, $+1.40$) and every cell of the bin-count sweep, the analyst choice with the largest leverage here (§S3.9, Table S7). On rows the search never saw the separation reproduces at about half its size. An independent ThermoML publication supplies 95 rows over 32 pairs the broad set does not carry, 21 solutes in diethylene and triethylene glycol at three temperatures. There the margin is $+1.06$ $[+0.74, +1.29]$, against the in-sample $+2.04$ $[+1.25, +2.87]$, and the two intervals overlap only in a narrow band. The ThermoML set carries one source publication, so the leave-one-source-out clause this section's rule turns on cannot be applied to it, and the reading is descriptive rather than admissible. It settles the sign. It suggests the in-sample magnitude is the upper end of what this chemistry supports rather than its size.

The sweep varies the binning bound's own settings and not the family of estimator.

Deviation from the declaration. A predeclared cross-fitted regressor was to replace the binning bound if it passed a validity gate, and on the declaration's own conservative source-grouped folds it does not return a bound at all. Lemma 1(a) puts B_{insuff} in $[0, \text{MSE}]$, and on the glycol-ether stratum itself the estimator returns $B_{\text{insuff}} = 4.92$ against that stratum's own MSE of 2.25, twice what the decomposition allows, at $R^2 = -0.60$ out of fold. Across the source-folded cells the same violation runs from 1.6 to 2.9 times MSE with a random forest and some twenty-seven orders of magnitude more with ridge. An upper bound outside the range its estimand can occupy is not a bound on it, so the -7.58 it prints here, and the loss of every admissible positive row set, are uninformative rather than reversals. The criterion has no free parameter and no direction: it fires whenever the output leaves the interval the decomposition allows. **The declared gate did not test it.** The gate read an out-of-fold R^2 and a pair-conditional variance threshold, and the permissive pair-grouped configuration passed both. **The declaration required unconditional adoption on that pass, and every margin printed in this paper is nevertheless the binning bound's.**

The cross-fit, where it does pass, is uniformly *looser*, and it does not favour us. On the glycol ethers it returns $+1.76$ $[+1.02, +2.49]$, so the sign holds. But the chemistry-blind null moves from 0.094 to 0.137. Conditioned on drawing a map that certifies anything at all, a relabelling reaches the observed maximum in 54% of draws against 16% under the binning bound (§S3.12; note *e* of Table S16). Under the estimator the declaration named, this cell sits at chance. That does not subtract from what is claimed, because nothing is claimed for it: it is reported as the survivor of a search and not as an established effect, and a survivor's frequency under a null is what it is under either estimator. The deviation therefore runs in this paper's favour and was chosen after the two bounds were compared, which is what a declaration exists to prevent. The rule was written against one hazard: taking the smaller bound cell by cell once the cells are seen. It did not anticipate that one valid bound would be uniformly looser than the other, where obeying it prints a bound that is valid and strictly less informative. Two things limit the deviation. It followed one global comparison of two bounds on one row set, not a per-cell selection. And the poorer cross-fitted map prints cell by cell in §S3.12, where the declared rule can be held against it. The sweep therefore establishes robustness within the binning bound, not across estimators.

The row set clears no multiplicity null, and that belongs beside the finding and not in a limitation: it is by construction the survivor of a 59-fold search. Our null relabels the chemistry blindly—each solvent and solute keeps its rows and takes another's class labels, and the stratum family is rebuilt and put through the same rule. Over 2000 such relabellings, on all 477 rows and at the cell the map deploys, its six admissible strata are above chance ($p = 0.01$) while its two admissible positive row sets sit at chance ($p = 0.286$). One row set or more is certified in 58% of draws. The size is the sharper statistic, and it leaves the finding where the counts do: a certified margin of $+2.04$ or more arises in 10.1% of all draws, 17% of those that certify anything. The instrument licenses the separation on the rows it was computed on. As a chemistry-specific effect it is unestablished against the search that found it. Each p is a frequency, not a bound in either direction, and the null measures two biases. Provenance survives the permutation in two halves that run opposite and stand un-netted, so the 10.1% cannot be read as conservative. The set taken whole also has a positive margin at this cell, so a randomly assembled stratum inherits one on average, inflating the 58% and not the 10.1%. The exactly matched null would permute chemistry and provenance jointly, which these data place out of reach, the two being confounded in them (§S3.6).

A selection objection is settled by the same measurement on rows the search never saw, so we ran one. The PGL 6th-edition IDAC database is another group's compilation, sixteen times the broad set's rows on a different profile database, scored by the same deployed closure (Table S27). The row set, the estimator cell, the admissibility rule and what would count as confirmation in each direction were fixed before any margin was computed (§S3.7). **The chemistry stayed untestable there**, for a reason about the data rather than the number: the cell the finding is about, a glycol-ether solvent with an *organic* solute, is empty there. *Not testable* is the third verdict the pre-declaration lists and counts as non-confirmation, and it follows from the row inventory alone. The pre-registration delivers the absence of confirming data here, not a failed replication. The nearest cell that database carries is water in glycol-ether and polyol solvents. The rule refuses it out of sample for the reason it refuses water-as-solute in sample, so its margin licenses nothing and the sign stays out of risk. Over that database's ten solvent classes, stated as description and not as a test (Table S11), every class either fails the rule or leaves the two terms merged. Every margin above the one admitted positive class sits on a class the rule refuses: out of sample the glycol ethers sit away from where this closure errs hardest.

The glycol-ether row set is the whole of what the instrument returned, and this work does not advance it as an established chemical effect. Reading it as "COSMO-SAC error exceeds input error" would carry it past five boundaries, each measured here. They are the map's other strata; both data sets taken whole; the solubility axis of §3.1, where the two terms stay merged; the kernel as against the composite it is deployed in; and the chemistry-blind null. The margin sits at the ninetieth percentile of that null, on the one chemistry for which out-of-sample rows are missing. What the second axis establishes is therefore the instrument and its map, and not this cell: a procedure that refuses fifty-eight strata of fifty-nine, and returns for the fifty-ninth a number it cannot separate from its own search.

What the other fifty-eight strata are. They are the rule refusing to answer, and the refusal is calibrated rather than severe. The chemistry-blind null above still certifies some row set in 58% of draws, so the rule accepts about as often on relabelled chemistry as on real. What refuses fifty-eight strata here is therefore the resolution of the data, not a threshold tuned until one cell survived. An instrument that returned a verdict on every stratum of a 477-row set would be reporting resolution it does not have, and the admissibility clauses exist to withhold exactly that. The 10.1% beside it is a different statistic about a different object. It is the frequency with which a relabelling reaches this stratum's own margin, and it is why the finding is stated as a search's survivor rather than as an established chemical effect. Of the six admissible cells, three are the glycol ethers at three granularities and the remaining three are two row sets. The alkane solutes are one of them, positive and demoted to the glycol ethers by clause (c); the acceptor-only aprotics are the other, entering at two granularities, and they pass the rule while leaving the two terms merged, the instrument being one-sided. The uniqueness is therefore narrower than being the sole admissible stratum, or the sole stratum sign-stable under every deletion: it is one row set admissible, positive and left standing by clause (c). §S3.8 records which cell each remaining stratum fails on, none of them used to explain a reversal.

Why the output is a map. $B^{\mathrm{up}}_{\mathrm{insuff}}$ is nearly the same in every stratum while MSE moves: a factor of 2.0 against 25 over the three classes the design can bound, and 6.8 against 51 once the three that carry an estimate are added. The 6.8 rests on a mono-alcohol figure the artifact flags as an invalid bound on a stratum already excluded. Against 2000 random partitions of the 477 rows into blocks of the nine observed class sizes, the chemical partition's range in MSE exceeds every draw while its range in the bound is unremarkable ($p = 0.26$). Their ratio gives $p = 0.02$ (§S3.8). The estimator is showing through. Each stratum is summarised by eight quantile bins of one scalar, so the bound measures the residual spread of m inside such a bin. The spread is a property of the binning and not of the chemistry, and the error moves independently of it. An aggregate margin is therefore a composition-weighted average of a term free to move 25-fold against one that barely moves. It reports a binding closure for any set whose composition includes a chemistry the closure fails on. The statement is about the aggregate, not about the bound, which stays valid in each stratum where it is computed.

3.5.2 The VT-2005-matched set (n=60)

This set carries the paper's other reference-matched analyses and is far weaker. All 60 rows sit at 298 K, so that $\dim(z^{\star})/n \approx 102/60$, and methanol occupies 25 of them. Its 17 solvents fall in six of the nine classes, each short of the n=40 a bound needs. The set supports no stratified bound, and we state no stratum of it as a finding. Its aggregate margin of +0.05 collapses four ways: under deletion of the two leverage pairs, under dropping methanol, under blocked out-of-fold estimators, and under 4 of the 60 single-pair deletions. The worst of those deletions leaves −0.03, where on the broad set every single-pair deletion leaves the sign intact. A margin here would in any case bound a label systematic tracking $z^{\star}$, which the unbiased within-bin variance leaves *open* (§S3.11). The reported cell is not a favourable one: drawn over the whole grid of bin counts and conventions the aggregate changes sign, and of the cells returning a positive margin the reported one is near the bottom (Fig. S2, §S3.3). Of the 60 pairs, 57 also appear in the broad IDAC set at 298 K under UD profiles, so the two are overlapping samples rather than independent ones. VT-2005, and therefore the paradox's own oracle, is defined on this sub-design, so the agreement between the two sets' error orderings is a consistency check rather than corroboration.

3.5.3 Whether a better closure moves the ceiling, and where this one fails

If the ceiling is the closure, the direct remedy is a better closure. The standard 2002 → 2010 step delivers one on the VT-2005-matched set and withholds it on the broad IDAC set. Neither half of that pair settles anything. The improvement is on a 57-pair overlap with the larger set rather than an independent sample, and the withholding is −0.99 at $P \approx 0.2$, which is not a measured equality either. Three further readings are of the same kind: the 2010 kernel on its own typed latent, a fitted kernel residual asked to transfer across solvents, and the block split that places the reversal in the chemistry rather than the temperature range. Each prints with its interval in Table S9 and §S6.6. We claim only that **the fidelity lever we can demonstrate is confined to the smaller set's chemistry**, and that broad-set null is itself a small-molecule and glycol-heavy one, so on drug-like solids the lever stands untested.

On the one row set the rule leaves standing, where B_{closure} dominates, the ceiling is in the map *as it is deployed*: the fixed kernel together with the one computed conformer it reads. The composite is what B_{closure} is the error of, and the object the pipeline runs on. The split between the two stays open here, and so does whether the composite is *fixably* wrong. A fitted kernel residual on the 60 matched pairs transfers within a solvent and fails to transfer across solvents, under the solvent-grouped folds that are the one fold design free of leakage. The result is a null. So B_{closure} is an error a handful of kernel parameters can absorb *within* a solvent, and its correctability across solvents we leave unshown.

The failure is chemically placed. On the VT-2005-matched set 90% of the squared error falls on strong hydrogen-bond-donor solvents, and the closure is near-exact on acceptor-only ones. Those are the chemistry the documented single-parameter H-bond term governs, and the term the donor/acceptor-typed 2010 closure was built to replace.[37,44] That failure is established already and the localisation restates it. The new part is that on one such chemistry the failure is bounded against what the closure's inputs leave unresolved. The bound also survives the obvious alternative: replacing the four glycol-ether references with Boltzmann ensembles moves the stratum's margin from +2.04 to +2.27, so the error sits in a *chemistry* rather than in the conformer half of the composite (§S3.10).

COSMO-SAC-2002 being a low-fidelity closure is the contribution and not a qualification of the result. It is what a practitioner adopts by default, and a closure's fidelity is *unknowable in advance* on a new system. The field routinely composes an expressive encoder with a fixed mechanistic map and assumes the map is good enough. Once that assumption fails, the learned latent silently departs from the quantity it is named after (§3.3), and an external oracle is what makes the departure visible. Outside that one row set the map reports cells it can only leave unresolved, which is a weaker statement than cells in which the closure is sound. The exposure the choice of profile database creates, and why that arithmetic sizes it while leaving its share unbounded, is in §S6.6.

3.6 The same split on a second closure: pKa through a fixed Hammett relation

The instrument generalises beyond COSMO-SAC, and a second domain runs it where the sign of grounding can be predicted before the measurement. The closure is a Hammett linear free-energy relationship $g(z^\star) = \mathrm{p}K_{a,0} - \rho\,\sigma^\star_{\mathrm{H}}$ in tabulated Hansch–Leo–Taft constants,[45] unfitted on these data, so g is fixed in the sense Lemma 1 requires. The learned quantity is the substituent constant predicted from structure, and the data are the OPERA compilation.[46] A rule reading structure alone splits the records into two poles. The *clean* pole is the 158 whose every substituted position is meta or para with a tabulated substituent. The *degraded* pole is the 846 where $\sigma^\star_{\mathrm{H}}$ is a partial sum over the positions the table can see.

The sign reverses between them, in the direction that construction fixes in advance. Across K=20 within-scaffold splits the reference *improves* accuracy on the clean pole (MAE 0.48 learned against 0.315, a gap of -0.165 at a 90% confidence interval of $[-0.28, -0.04]$) and *degrades* it on the degraded pole (0.83 against 1.62, $+0.79$ $[+0.57, +0.95]$). The freely learned constant absorbs what the zeroed positions carry and the tabulated one lacks. **The flip is an in-distribution result and the scaffold protocol does not certify it.** Re-run on a scaffold split the reference helps on both poles, so the degraded pole's "grounding hurts" half fails to certify and so does the reversal. The scaffold run confirms the mechanism, not the flip. We measure every solubility result in this paper on a scaffold split, and this is the one place the protocol changes. §S6.10 carries the assignment rule, the bounds on both poles, what the quality filter decides, the estimator's two conditions of use, the competence sweep and the G=4 sign test.

3.7 Scope, accuracy, and limitations

Two measurements, in different places. On solubility, supervising a learned physical latent on reference values during training moves the error without an established sign, while *supplying* those values at inference hurts at every seed. On infinite-dilution activity data, and there on one class of solvent, the fixed map's own error exceeds the bound on what its inputs leave undetermined. The two sit in different regimes on largely disjoint chemistry, so "the substitution hurts *because* the map is misspecified" remains untested (limitation (i)). The finding is one-sided: one row set of the map on which grounding effort is *not* the remedy.

The unconstrained control's representation. If the physics arm's intermediate is a surrogate, the thermodynamics might still be present implicitly in the control, which is free to represent it. It is absent: a ridge probe of the control's pair representation recovers the model's own prediction at $R^2 = 0.98$ and leaves the measured ideal-solubility term unrecovered. The negative falls short of an exclusion. It rests on 13 held-out solute clusters whose reference scarcity caps them, each measured R^2 sits at or below its own permutation null's ceiling, and no power calculation says how far above that ceiling the design reaches. It is limitation (ix), and §S9 carries the three studies, the controls and the temperature probe.

Where the accuracy stands, and the confound in it. None of this work's arms is competitive with current solubility models, and none is offered as one; the comparison in Table S30 fixes the scale this work operates on and does not rank anything. The substitution result is a contrast within one checkpoint and does not depend on that scale. The physics arms also trail their own control in the seed mean (Table S4), but the two sides are separately tuned with no hyperparameter-matched control, so no difference between them, and none of the chemical strata that resolve the same difference (§S6.4, §S6.5), is attributable to the thermodynamic bottleneck. We claim no accuracy result in either direction, and the matched control that would repair this is unrun. Untouched by that confound is every *within-checkpoint* contrast, where the two sides share one trained model and one tuning (§3.1, §3.2, §3.3, §3.5). Two fall outside it, comparing across trainings or across the separately tuned control: the supervision gain and §3.2's shrinkage slopes. Grounding the crystal factor on external labels is underpowered rather than null. 97.76% of that pool was melting points the model had already been trained on, and where it carried something new the crystal head moved. On the held-out solutes it is the worse of the two, and the downstream solubility gap crosses zero once clustered by solute (§S6.3). Temperature extrapolation is the one place the physical *structure* helps, and it helps as a property of the hypothesis class and not of any arm trained here. Fitting below $T \leq 309.75$ K and scoring at $T \geq 330$ K on the same 1751 solute–solvent pairs withholds temperature and keeps structure. On that split a structured van't Hoff class reaches an $\ln x_2$ MAE of 0.37 against 1.29 for a random forest over Morgan fingerprints and temperature, each one fit on one temperature cut with no replicate behind it (§S6.1).

The external block is on an easier by-solute split, and the harness for a like-for-like comparison is deposited and unrun. The difference that remains is a scale rather than a placement. This work's control reaches $\log_{10} S$ RMSE 1.00 on held-out scaffolds and its physics arms 1.44 to 1.64, against 0.83 to 0.99 for the two leading direct models and 1.43 to 2.16 for the physics-informed SolProp cycle.[2] All six are as published by Attia et al.,[1] in $\log_{10} S$ RMSE in $\mathrm{mol\,L^{-1}}$, on two other corpora: the SolProp test set, whose solutes are disjoint from its training corpus, and the near-ambient Leeds set of Boobier et al.[47] Sharing no row with these, they set a scale and withhold an ordering; each range is two point values on two corpora rather than a spread, and none of the six carries a published uncertainty. The clip sweep, the random-pair rows and the conversion behind the $\log_{10} S$ column are in §S4.

These negatives fall where Remark 1 predicts a prior will hurt (§2.1.2). The synthetic sweep of Fig. S10 is true by construction there, so it verifies the instrument and not the hypothesis; the pKa closure exercises the sufficiency channel on real data.

Limitations. All fourteen, keeping the numerals used throughout; §S6.8 carries the qualifying detail for the nine it holds.

(i) The phenomenon and its causal attribution lie in *different regimes*. The paradox is where the model operates (finite composition, all γ, n=5608). The attribution and the latent-departure result rest on infinite-dilution activity data alone (n=477 across temperatures, and the 298 K, low-γ, VT-2005-matched n=60 and n=44 sets). Transfer back to the paradox's own regime is a conjecture and the principal open question here. The n=60 set is all gas chromatography and all from three publications, 33 pairs from one, so a method-stratified or source-clustered re-estimate lies beyond it. *This set's* aggregate margin inverts under a broader ThermoML pull where the broad set holds steady. It is a different object from the glycol-ether stratum of limitation (vi), which lives on the broad set and whose own ThermoML test reproduces. Both sets sit outside the deployment regime (§S8). No set carries both quantities, and none is waiting to be assembled: across the labelled corpus's 12,129 solute–solvent pairs and the 3,145 of the largest infinite-dilution pull we hold, not one pair carries a solubility and a γ^∞ alike, and the 22 solvents and 9 solutes the two sides share still meet in no common pair.

(ii) What the convention rule costs the aggregate (§2.5).

(iii) That other cells of the map keep their sign as well (§3.5.1).

(iv) B_{insuff} is upper-bounded rather than point-identified, the design holding no replicates at fixed $z^\star$. B_{closure} is the error of a composite, the fixed map together with the calculation that produced $z^\star$, and the measurements here leave it undivided between the two and its share unbounded (§3.5.3).

(v) The constant-offset bound's convention-specificity (§2.5).

(vi) The stratification reads structure alone, so every cell was fixed before its outcome was seen. Cells from different axes share rows and are therefore dependent evidence, counted once; granularity is an analyst choice, which is why we report three solvent levels and two solute levels. The per-cell bootstrap frequencies carry no multiplicity control, so one cell's P describes that cell alone, and the admissibility rule is a stability filter rather than a correction for the search. Neither test of the search establishes a chemical effect: under the chemistry-blind relabelling null the shape of the finding actually claimed sits at chance (§3.5.1), and the pre-declared out-of-sample test on the PGL set returned *not testable*. A second pre-declared test, on 95 ThermoML rows the search never saw, does return a positive margin at about half the in-sample size (§3.5.1). It rests on one source publication, so it settles the sign without meeting this rule's own stability clause. We therefore do not advance that cell as a chemical result at all. It is reported as what the instrument returned on its own rows, and the same holds a fortiori for every cell the map reports and leaves unstated. With B_{insuff} bounded from above alone, every cell is also silent on whether better inputs would help there, so the map's one action is to withhold a chemistry from an allocation of grounding effort.

(vii) The accuracy ceiling standing a factor of five above the label-noise floor (§2.2).

(viii) What the three-seed latent-departure result establishes and where it stops (§3.3).

(ix) What bounds the black-box probe (the probe paragraph above).

(x) What the closure-fidelity contrast leaves unmeasured, and what would settle it (§3.4).

(xi) No accuracy difference between the arms is attributable to the bottleneck (above). Two axes shape whether grounding helps, closure fidelity and how the latent is trained, and this work leaves that plane uncharacterised on real systems. The chemical strata form a pattern in the direction the sign rule would give, but separately tuned pipelines can produce one and a multiplicity control is missing. §S6.4 therefore records it as the experiment a matched control should run, and the diagnostic is *proposed* and not validated.

(xii) The scaffold-leak guard is a property of the released stream builder rather than a certified property of the runs §3.1 reports. The re-run that *is* certified leak-free returns the supervision gain without a sign, so no reported number now rests on the uncertified build. The superseded three-seed gain of 0.20 was the one that did, and deleting the eight reachable solutes left it at $+0.17$ $[+0.03, +0.31]$. The deletion bounds where a leak could have acted, as do the arguments that bound the other contrasts (§2.2).

(xiii) What the deposit leaves uncheckable (the data-availability statement).

(xiv) One trained COSMO-SAC head survives in the deposit, and it is a poor model: a retired run at MAE 2.61 and $R^2 = -0.31$

on the scaffold split, worse than predicting the corpus mean. Four quantities are read off it, each stated with that scope where it prints: the donor-window areas of Fig. 4, the cavity-area ratios beside them, the substitution penalty's movement with the segment count, and the size of the substitution step against a random direction of the same norm. Each is a property of a σ-profile and its closure rather than of an arm's accuracy, and none of them enters a reported margin; the donor-window occupancy is additionally reproduced by an *untrained* network of the same architecture, so it prices the parameterisation and not the training. What a better checkpoint would change is unmeasured, and the arms of Fig. 2 retain no weights to supply one.

(xv) Two terms the thermodynamics asks for are set to zero in the deployed model, and both are absorbed at training time by the same learned σ head. The first is ΔC_p in the ideal term, one-signed and worth a median 0.32 in $\ln x_2$ where a melting point is measured (§2.1). The second is the Staverman–Guggenheim combinatorial term, off in every arm the paradox is reported on. Both cancel in the evaluation-time substitution and both survive the training that produced the drift of §3.3, so the mechanism there reaches beyond the activity kernel.

4 Conclusions

Grounding a learned physical intermediate in reference values is two operations rather than one, and here only one of them carried a sign. Substituting the reference profile for the learned one at prediction time raised solubility error at every one of five leak-free seeds, and moved the ranking of solvents for a solute against the practitioner at every one of them. Supervising the same head against the same database during training moved the error the other way at four seeds of five. The operation that failed is the one a practitioner reaches for, because it is the one that runs without retraining. That asymmetry is what this work establishes.

A prediction generalises from this, rather than a design rule. Wherever a predictor composes a fixed physical model over a learned intermediate for which an external reference exists, that reference measures whether the fixed model itself limits accuracy. Where it answers, the answer decides the remedy: a more faithful physical model, the calculation that supplies its reference values included. The answer comes chemistry by chemistry rather than once for a field, and it comes one-sided, returning the chemistry on which better inputs are not the remedy. Our fixed model is COSMO-SAC at its 2002 parameterisation, whose kernel cannot move, so a sign that turned on kernel mobility would lie outside what these measurements reach.[15] The experiment this work leaves is to *measure* γ^∞ on pairs that already carry a solubility, because the two literatures share no pair and no wider pull will supply one.

Three things follow for anyone building on this. Measure the two quantities on the same pairs: the bridge between the regimes is a laboratory programme rather than a data-assembly task, and it is what would turn the mechanism here into an explanation. Declare the estimator cell before the margin, and where two valid bounds are available fix the rule for choosing between them in the same document rather than after seeing both — this work's own departure from that is recorded in §3.5.1 and is the sharpest thing a reader can hold against it. And if a learned intermediate must feed a fixed physical model, freeze it against its reference and fine-tune only downstream: the one published construction that does so reports the opposite sign to ours,[15] and that difference in training design is the most actionable contrast here.

Author contributions

N. L. Polomoshnov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. A. V. Rudik: Writing – review

& editing.

Conflicts of interest

There are no conflicts to declare.

Data and Software Availability

The analysis code, and those intermediate artifacts small enough to version, are available at `https://github.com/doctawho42/tgnn-solv` (commit **[hash to be supplied at submission]**) under the MIT licence, that commit being the one the reported numbers were produced from. The model checkpoints, the processed split, the full per-arm prediction files and the larger analysis artifacts will be deposited in a Zenodo archive, **[whose DOI is registered at submission]**. The data sources are public. Solubility is BigSolDB 2.0;[33] the reference σ-profiles are the VT-2005 database, and the UD profiles Ref.[44] as redistributed with Ref.;[36] the infinite-dilution activity coefficients are ThermoML, with the PGL 6th-edition IDAC database behind Table S27; the pKa data are the OPERA compilation.[46] Four disclosed defects sit on the deposited record, in §S8 and §S3. The three-seed surrogate run's per-run split provenance was not logged; one σ-stream build on disk was written against the wrong split files; the two VT-2005-matched audit artifacts disagree on one out-of-fold estimate. The fourth is that the σ-stream those runs consumed is recorded by digest in every checkpoint manifest and was not itself retained. Rebuilding it from the recorded parameters reproduces the pool exactly, the same 1319 molecules with bit-equal profiles, and the train/validation split sizes exactly, but no retained file settles the assignment between them. That comparison is itself deposited, digest by digest, and regenerates from the released code. Every number reported here was measured on arms whose weights, predictions and manifests are deposited; what the loss bounds is what can be added to them, since the validation half early-stops the σ warm-up. Two further defects are repaired rather than disclosed, each described where its numbers are printed. We re-scored the two FastSolv retraining bundles (Tables S30 and S20, with the diagnosis, the withdrawn values and the control that reproduces them in §S4), and the withdrawn bundle is kept beside the re-scored one. And the truncated seed-44 σ-oracle per-row file has been recovered complete and now stands in the deposit (Table S29, §S8). *The audit in one place.* Twelve defects and limits are disclosed in this submission, each described where its numbers print, and they are of five kinds. Four are properties of the deposit and are disclosed rather than repaired: an unlogged split provenance, a stream build written against the wrong split files, two audit artifacts disagreeing on one out-of-fold estimate, and the σ-stream that was not retained. Two were errors in this work's own numbers and are repaired, with the withdrawn values kept beside the corrected ones: the twice-scaled temperature in the FastSolv retraining bundles, and the truncated seed-44 per-row file, now recovered complete. Three are cases where a check this work built refuted or weakened something this work had claimed: the departure from the estimator declaration (§3.5.1), the linear probe whose verdict an untrained encoder of the same architecture also earns, and the donor-window occupancy that an untrained network reaches as fully as the trained one (§3.3). Two are properties of the model rather than of a number, disclosed and then bounded: the activity head violates the identity on a self-pair, which no scored row is (§2.4), and a deposited out-of-fold forest bound is invalid on row-level folds, which the artifact says in its own file. One is a limit of the checking apparatus itself, disclosed beside what it catches: value-conservation checks do not detect loss of *role* (§S8). None of the twelve is reported by anyone but us, and each is stated where it bites rather than collected into a caveat.

What that deposit does and does not make checkable is limitation (xiii). Several of the numbers the verdict rests on are checkable by re-running the released code and by that route alone, and its input artifacts are larger than the repository carries. We therefore deposit the two decomposition sets as row-level tables (Data Sets S1 and S2), so those sets and their margins can be recomputed with no code at all. That leaves the provenance of the closure evaluations inside them, and of the trained arms, undelivered.

Acknowledgement This research received no external funding. Computations were carried out on commercial cloud services (Google Colab Pro+, Modal, and Google Cloud Platform) using NVIDIA T4 and L4 GPUs. Generative-AI use: the authors used Anthropic's Claude for analysis and verification code, for the figures that code generates, and for drafting and editing. It generated and altered no measurement: every number in both documents is produced by a deposited script from a deposited artifact, and the checks of §S8 bind them there. The authors are responsible for all of the content.

Supporting Information Available

The Supporting Information is a separate document. Its sections, tables, figures and equations are numbered S1, S2, …, and every reference above that carries an S is to it. It contains supplementary methods; the proofs of Lemma 1 and Remark 1 and three further results; per-arm tables and decomposition robustness; the map's 59 strata in full, including the 44 the estimator cannot bound. It also contains further external ablations; supplementary mechanism results; supplementary diagnostics and broad negatives; the second-domain pKa construction; data provenance and reproducibility; the black-box probe tables; the synthetic closure-fidelity family; and the positioning against adjacent literatures. We deposit two machine-readable tables with it: the 477-row broad IDAC set (Data Set S1) and the 60-row VT-2005-matched set (Data Set S2), and their columns are listed in §S8.

References

(1) Attia, L.; Burns, J. W.; Doyle, P. S.; Green, W. H. Data-driven organic solubility prediction at the limit of aleatoric uncertainty. *Nature Communications* **2025**, *16*, 7497, FASTSOLV model; earlier ChemRxiv 2024 preprint doi:10.26434/chemrxiv-2024-93qp3.

(2) Vermeire, F. H.; Chung, Y.; Green, W. H. Predicting Solubility Limits of Organic Solutes for a Wide Range of Solvents and Temperatures. *Journal of the American Chemical Society* **2022**, *144*, 10785–10797.

(3) Winter, B.; Winter, C.; Schilling, J.; Bardow, A. A smile is all you need: Predicting limiting activity coefficients from SMILES with natural language processing. *Digital Discovery* **2022**, *1*, 859–869.

(4) Sanchez Medina, E. I.; Linke, S.; Stoll, M.; Sundmacher, K. Gibbs–Helmholtz graph neural network: capturing the temperature dependency of activity coefficients at infinite dilution. *Digital Discovery* **2023**, *2*, 781–798.

(5) Wang, J.; Ye, J.; Zhang, W. GHGEAT: Gibbs–Helmholtz-Constrained Graph External Attention for Temperature-Dependent Activity Coefficient Prediction at Infinite Dilution. *Journal of Chemical Information and Modeling* **2026**, *66*, 3759–3773.

(6) Lin, S.-T.; Sandler, S. I. A Priori Phase Equilibrium Prediction from a Segment Contribution Solvation Model. *Industrial & Engineering Chemistry Research* **2002**, *41*, 899–913.

(7) Mullins, E.; Oldland, R.; Liu, Y. A.; Wang, S.; Sandler, S. I.; Chen, C.-C.; Zwolak, M.; Seavey, K. C. Sigma-Profile Database for Using COSMO-Based Thermodynamic Methods. *Industrial & Engineering Chemistry Research* **2006**, *45*, 4389–4415.

(8) Mu, T.; Rarey, J.; Gmehling, J. Group contribution prediction of surface charge density profiles for COSMO-RS(Ol). *AIChE Journal* **2007**, *53*, 3231–3240.

(9) Mu, T.; Rarey, J.; Gmehling, J. Group contribution prediction of surface charge density distribution of molecules for COSMO-SAC. *AIChE Journal* **2009**, *55*, 3298–3300.

(10) Kang, J.-L.; Chiu, C.-T.; Huang, J. S.; Wong, D. S.-H. A surrogate model of sigma profile and COSMOSAC activity coefficient predictions using transformer with SMILES input. *Digital Chemical Engineering* **2022**, *2*, 100016.

(11) Chen, G.; Song, Z.; Qi, Z. Transformer-convolutional neural network for surface charge density profile prediction: Enabling high-throughput solvent screening. *Chemical Engineering Science* **2021**, *246*, 117002.

(12) Zhang, J.; Wang, Q.; Shen, W. Message-passing neural network based multi-task deep-learning framework for COSMO-SAC based sigma-profile and VCOSMO prediction. *Chemical Engineering Science* **2022**, *254*, 117624.

(13) Salih, F. Y. M.; Abranches, D. O.; Maginn, E. J.; Colón, Y. J. Open-source generation of sigma profiles: impact of quantum chemistry and solvation treatment on machine learning performance. *Digital Discovery* **2025**, *4*, 2711–2723.

(14) Islam, M. R.; Chen, C.-C. COSMO-SAC Sigma Profile Generation with Conceptual Segment Concept. *Industrial & Engineering Chemistry Research* **2015**, *54*, 4441–4454.

(15) Yang, Y.; Lin, S.-T. Physics-Embedded Machine Learning Model for Phase Equilibrium Prediction in Multicomponent Systems. *Journal of Chemical Information and Modeling* **2025**, *65*, 10180–10193.

(16) Specht, T.; Nagda, M.; Fellenz, S.; Mandt, S.; Hasse, H.; Jirasek, F. HANNA: hard-constraint neural network for consistent activity coefficient prediction. *Chemical Science* **2024**, *15*, 19777–19786.

(17) Karniadakis, G. E.; Kevrekidis, I. G.; Lu, L.; Perdikaris, P.; Wang, S.; Yang, L. Physics-informed machine learning. *Nature Reviews Physics* **2021**, *3*, 422–440.

(18) Raissi, M.; Perdikaris, P.; Karniadakis, G. E. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics* **2019**, *378*, 686–707.

(19) Willard, J.; Jia, X.; Xu, S.; Steinbach, M.; Kumar, V. Integrating Scientific Knowledge with Machine Learning for Engineering and Environmental Systems. *ACM Computing Surveys* **2022**, *55*, 1–37.

(20) Ma, L.; Boullé, N.; Yang, Y.-S.; Wu, H.; Guo, L. Physics-Guided Correction for Operator Learning under Model Misspecification. *arXiv preprint arXiv:2606.03469* **2026**,

(21) Krishnapriyan, A. S.; Gholami, A.; Zhe, S.; Kirby, R. M.; Mahoney, M. W. Characterizing possible failure modes in physics-informed neural networks. Advances in Neural Information Processing Systems (NeurIPS). 2021.

(22) Mohan, A.; Chattopadhyay, A.; Miller, J. What You See is Not What You Get: Neural Partial Differential Equations and The Illusion of Learning. *arXiv preprint arXiv:2411.15101* **2024**,

(23) Koh, P. W.; Nguyen, T.; Tang, Y. S.; Mussmann, S.; Pierson, E.; Kim, B.; Liang, P. Concept Bottleneck Models. Proceedings of the 37th International Conference on Machine Learning. 2020; pp 5338–5348.

(24) Mahinpei, A.; Clark, J.; Lage, I.; Doshi-Velez, F.; Pan, W. Promises and Pitfalls of Black-Box Concept Learning Models. *arXiv preprint arXiv:2106.13314* **2021**,

(25) Margeloiu, A.; Ashman, M.; Bhatt, U.; Chen, Y.; Jamnik, M.; Weller, A. Do Concept Bottleneck Models Learn as Intended? *arXiv preprint arXiv:2105.04289* **2021**,

(26) Havasi, M.; Parbhoo, S.; Doshi-Velez, F. Addressing Leakage in Concept Bottleneck Models. Advances in Neural Information Processing Systems (NeurIPS). 2022.

(27) Parisini, E.; Chakraborti, T.; Harbron, C.; MacArthur, B. D.; Banerji, C. R. S. Leakage and Interpretability in Concept-Based Models. *arXiv preprint arXiv:2504.14094* **2025**,

(28) Espinosa Zarlenga, M.; Dominici, G.; Barbiero, P.; Shams, Z.; Jamnik, M. Avoiding Leakage Poisoning: Concept Interventions Under Distribution Shifts. *arXiv preprint arXiv:2504.17921* **2025**,

(29) Kennedy, M. C.; O'Hagan, A. Bayesian Calibration of Computer Models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* **2001**, *63*, 425–464.

(30) Brynjarsdóttir, J.; O'Hagan, A. Learning about physical parameters: the importance of model discrepancy. *Inverse Problems* **2014**, *30*, 114007.

(31) Hirano, K.; Imbens, G. W.; Ridder, G. Efficient Estimation of Average Treatment Effects Using the Estimated Propensity Score. *Econometrica* **2003**, *71*, 1161–1189.

(32) Robins, J. M.; Ritov, Y. Toward a Curse of Dimensionality Appropriate (CODA) Asymptotic Theory for Semi-Parametric Models. *Statistics in Medicine* **1997**, *16*, 285–319.

(33) Krasnov, L.; Malikov, D.; Kiseleva, M.; Tatarin, S.; Sosnin, S.; Bezzubov, S. BigSolDB 2.0, dataset of solubility values for organic compounds in different solvents at various temperatures. *Scientific Data* **2025**, *12*, 1236.

(34) Frenkel, M.; Chirico, R. D.; Diky, V. V.; Yan, X.; Dong, Q.; Muzny, C. ThermoML: An XML-Based Approach for Storage and Exchange of Experimental and Critically Evaluated Thermophysical and Thermochemical Property Data. *Journal of Chemical Information and Computer Sciences* **2003**, *43*, 1344–1359.

(35) Ramani, V.; Arora, H. A.; Kuchhal, D.; Tatarin, S.; Krasnov, L.; Ranu, S.; Karmakar, T. SC3: The Multi-Solvent Solubility Challenge and Benchmark. 2026.

(36) Bell, I. H.; Mickoleit, E.; Hsieh, C.-M.; Lin, S.-T.; Vrabec, J.; Breitkopf, C.; Jäger, A. A Benchmark Open-Source Implementation of COSMO-SAC. *Journal of Chemical Theory and Computation* **2020**, *16*, 2635–2646.

(37) Hsieh, C.-M.; Sandler, S. I.; Lin, S.-T. Improvements of COSMO-SAC for Vapor–Liquid and Liquid–Liquid Equilibrium Predictions. *Fluid Phase Equilibria* **2010**, *297*, 90–97.

(38) Hsieh, C.-M.; Lin, S.-T.; Vrabec, J. Considering the Dispersive Interactions in the COSMO-SAC Model for More Accurate Predictions of Fluid Phase Behavior. *Fluid Phase Equilibria* **2014**, *367*, 109–116.

(39) de Souza, E. T., Jr.; Alcantara, M. L.; Staudt, P. B.; Coutinho, J. a. A. P.; Soares, R. d. P. Development of a COSMO-SAC Parametrization with Advanced QM Method TZVPD-FINE. *Industrial & Engineering Chemistry Research* **2025**, *64*, 14700–14711.

(40) Nael, M. A.; Alakonda, L. M.; Elokely, K. M. Defining the Data set Defines the QSAR Claim. *Journal of*

Chemical Information and Modeling **2026**, *66*, 2951–2954.

(41) Takeishi, N.; Kalousis, A. Physics-Integrated Variational Autoencoders for Robust and Interpretable Generative Modeling. Advances in Neural Information Processing Systems (NeurIPS). 2021.

(42) Internò, C.; Yamaguchi, J.; Amdahl-Culleton, L.; Olhofer, M.; Klindt, D.; Hammer, B. The Observer Effect in World Models: Invasive Adaptation Corrupts Latent Physics. *arXiv preprint arXiv:2602.12218* **2026**,

(43) Wu, J.; Xiao, H.; Sun, R.; Wang, Q. Reynolds-averaged Navier–Stokes equations with explicit data-driven Reynolds stress closure can be ill-conditioned. *Journal of Fluid Mechanics* **2019**, *869*, 553–586.

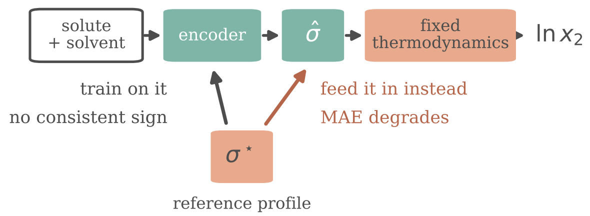
(44) Fingerhut, R.; Chen, W.-L.; Schedemann, A.; Cordes, W.; Rarey, J.; Hsieh, C.-M.; Vrabec, J.; Lin, S.-T. Comprehensive Assessment of COSMO-SAC Models for Predictions of Fluid-Phase Equilibria. *Industrial & Engineering Chemistry Research* **2017**, *56*, 9868–9884.

(45) Hansch, C.; Leo, A.; Taft, R. W. A Survey of Hammett Substituent Constants and Resonance and Field Parameters. *Chemical Reviews* **1991**, *91*, 165–195.

(46) Mansouri, K.; Cariello, N. F.; Korotcov, A.; Tkachenko, V.; Grulke, C. M.; Sprankle, C. S.; Allen, D.; Casey, W. M.; Kleinstreuer, N. C.; Williams, A. J. Open-source QSAR models for pKa prediction using multiple machine learning approaches. *Journal of Cheminformatics* **2019**, *11*, 60.

(47) Boobier, S.; Hose, D. R. J.; Blacker, A. J.; Nguyen, B. N. Machine learning with physicochemical relationships: solubility prediction in organic solvents and water. *Nature Communications* **2020**, *11*, 5753.

TOC Graphic



Grounding is two operations on one database; only one of them has a sign (solubility MAE on held-out scaffolds, five seeds; magnitudes in the text).